

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E-30004-AK.pvdb

Date of Analysis : Aug 12, 2026 10:12am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1400.000 mm ID
 Vessel Design Length, Tangent to Tangent 908.40 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 93 °C
 Channel Side Design Temperature 66 °C
 Shell Side Design Pressure 157.489 kgf/cm²
 Channel Side Design Pressure 121.145 kgf/cm²
 Shop Shell Side Test Pressure 196.957 kgf/cm²
 Shop Channel Side Test Pressure 152.753 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Flange	SA-516 70	...	...	K02700	+IT
Flange	SA-266 2	...	...	K03506	+IT
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-266 2	...	...	K03506	
Tubes		...	...		
Tubesheet	SA-266 2	...	...	K03506	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Ch Cover	121.339	0.00	130.000	3.1750	N/A	No
LHS Ch flg	121.284	0.00	124.000	3.1750	N/A	No
ch shell	121.284	0.00	124.000	3.1750	N/A	No
Main Shell	157.628	0.00	160.000	3.1750	N/A	No
Dish End	157.626	0.00	158.000	3.1750	No	No

Liquid Level: 195.00 cm Dens.: 992.245 kg/m³ Sp. Gr.: 0.993

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	31.8	31.5	323.0	315.0	304.3	...	1.00	1.00
Body Flg	71.3	39.5	273.0	265.0	259.5	...	1.00	1.00
Cylinder	221.3	150.0	68.0	68.0	66.5	...	1.00	1.00



Cylinder	903.3	650.0	88.0	88.0	86.5	...	1.00	1.00
Ellipse	908.4	5.1	90.0	76.0	76.0	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube inlet	WRC-297	0.979	Shell	Passed
N2-Tube Outlet	WRC-297	0.979	Shell	Passed
N3-Shell Inlet	WRC-297	0.987	Shell	Passed
N4-Shell Outle	WRC-297	0.986	Shell	Passed

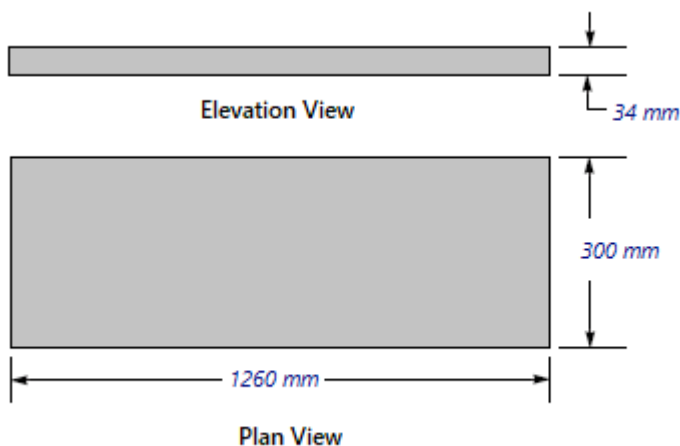
Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1200.000	mm
Wear Pad Width	600.000	mm
Wear Pad Thickness	16.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1625.000	mm

Baseplate Length	1260.000	mm
Baseplate Thickness	34.000	mm
Baseplate Width	300.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	500.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Maximum Tensile Bolt Load 1306. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	94489.70	kgf
Maximum Transverse Saddle Shear Load	5197.48	kgf
Maximum Longitudinal Saddle Shear Load	10394.96	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	78437.23	kgf
Maximum Transverse Saddle Shear Load	497.99	kgf
Maximum Longitudinal Saddle Shear Load	194.22	kgf

Weights:

Fabricated - Bare w/o removable internals 69856.0 kg



Shop Test	- Fabricated + water (full)	80914.0	kg
Shipping	- Fab. + rem. intls.+ shipping app.	69992.5	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	69992.5	kg
Empty	- Fab. + intls. + details + wghts.	69992.5	kg
Operating	- Empty + operating liquid (no ca)	80969.5	kg
Field Test	- Empty weight + test medium (full)	72161.7	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N1-Tube inlet	122	...	...	OK	OK	Passed
N2-Tube Outlet	122	...	...	OK	OK	Passed
N3-Shell Inlet	160	...	...	OK	OK	Passed
N4-Shell Outlet	160	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 160 Nozzle : N3-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 122 Nozzle : N2-Tube Outlet [Tubside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	N1-Tube inlet	1463.175	90.000	165.913
30	N2-Tube Outlet	1463.175	270.000	165.913
40	N3-Shell Inlet	3233.175	270.000	165.913
40	N4-Shell Outlet	7533.175	90.000	165.913

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube inlet	400 mm	160	WNF	406.4	40.462	606.40	68.00	398	900
N2-Tube Outlet	400 mm	160	WNF	406.4	40.462	606.40	68.00	398	900
N3-Shell Inlet	400 mm	160	WNF	406.4	40.462	...	...	466	1500
N4-Shell Outlet	400 mm	160	WNF	406.4	40.462	...	...	466	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube inl	SA-106 B	68.000	10.000	68.000	68.000	...
N2-Tube Out	SA-106 B	68.000	10.000	68.000	68.000	...
N3-Shell In	SA-266 2	88.000	10.000	...	...	...
N4-Shell Ou	SA-266 2	88.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube inlet	146.317	90.0	300.00	0.00	ch shell
N2-Tube Outlet	146.317	270.0	300.00	0.00	ch shell
N3-Shell Inlet	323.318	270.0	300.00	0.00	Main Shell
N4-Shell Outlet	753.318	90.0	300.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 323.0mm THK X 1960.0mm OD	SA-516 70
1	WELD NECK FLANGE: 273.0mm THK X 1960.0mm OD	SA-266 2
1	ELLIPTICAL HEAD: 2.0 X 1, 90.0mm THK X 1400.0mm ID X 5.1cm	SA-516 70
1	TUBESHEET: 1576.0mm X 320.0mm THK	SA-266 2
2298	TUBES: 6096.0mm X 19.0mm DIA X 3.0mm THK	SA-179
2	SADDLE: 200.0mm X 120 DEG	SA-516 70
1	INSULATION: 6500mm X 25mm THK	Hot
1	INSULATION: 395mm X 25mm THK	Hot
1	GASKET: 1433.0mm OD X 1407.0mm ID	...
32	BODY FLANGE BOLTS: 82.6mm DIA	SA-193 B7
64	NUTS FOR BODY FLANGE BOLTS: 82.6mm DIA	...
1	CYLINDRICAL SEGMENT 68.0mm THK X 3048.0mm X 150.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 68.0mm THK X 1350.2mm X 150.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 88.0mm THK X 3048.0mm X 650.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 88.0mm THK X 1350.2mm X 650.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Main Shell [8]	!	-29		Yes			
Dish End [10]	!	-29		Yes			
Dish End [7]	!	-29		Yes			
Left Saddle [24]	B	-13	-13	Yes	1.000	16.000	1.00
New Saddle [24]	B	-13	-13	Yes	1.000	16.000	1.00
N3-Shell Inlet[1]	!	3	-20	Yes	0.610	35.404	1.00
Nozzle Flg [5]	B	-18	-40	Yes			
N4-Shell Outle[1]	!	3	-21	Yes	0.610	35.404	1.00
Nozzle Flg [5]	B	-18	-40	Yes			
Tubesheet: SS[13]	!	-29		Yes			
Warmest MDMT:		3	-13				
LHS Ch Cover [11]	!	-29		No			
LHS Ch flg [11]	!	-29		Yes			
ch shell [8]	!	-29		Yes			
N1-Tube inlet [1]	B	14	13	Yes	0.975	68.000	1.00
Nozzle Flg [5]	A	-18	-30	Yes			
N2-Tube Outlet[1]	B	14	13	Yes	0.976	68.000	1.00
Nozzle Flg [5]	A	-18	-30	Yes			
Tubesheet: CS[14]	!	-29		Yes			
Bolting [21]		-40					
Warmest MDMT:		14	13				
Exchanger Side		Computed MDMT	Required MDMT	Pass/Fail			
Shell		-13.0	-28.9	Fail			
Channel/Tube		13.0	-28.9	Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

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Units used in this Analysis (KgfcM2):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	157.49	kgf/cm ²
Channel Side Design Pressure	121.15	kgf/cm ²
Shell Side Design Temperature	93.3	°C
Channel Side Design Temperature	65.6	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	Pneumatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	350	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM	
Importance Factor for IS-1893	2.000	
Soil Factor	1.000	
Zone Number	2.000	
Percent Seismic for Hydrotest	0.000	
Period of Vibration (User defined) Sec	0.000	
Damping Factor	2.000	



User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Ch Cover
Distance "FROM" to "TO"	31.5 cm
Flange Outside Diameter	1960 mm
Element Thickness	315 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	323 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	121.15 kgf/cm ²
Design Temperature Internal Pressure	66 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	66 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³



P Number Thickness	0	mm
Yield Stress, Operating	2507.2	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	195	cm
Liquid Density	992.24	kg/m ³

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Flange	
Description	LHS Ch flg	
Distance "FROM" to "TO"	39.5	cm
Flange Inside Diameter	1400	mm
Element Thickness	265	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	273	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	121.15	kgf/cm ²
Design Temperature Internal Pressure	66	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	66	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2 +IT	
Allowable Stress, Ambient	1682.4	kgf/cm ²
Allowable Stress, Operating	1589.9	kgf/cm ²
Allowable Stress, Hydrotest	2421.7	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2374.8	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	140	cm
Liquid Density	992.24	kg/m ³

Element#	3/5	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	ch shell	
Distance "FROM" to "TO"	150	cm
Inside Diameter	1400	mm
Element Thickness	68	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	121.15	kgf/cm ²
Design Temperature Internal Pressure	66	°C



Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	66	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2507.2	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	140	cm
Liquid Density	992.24	kg/m ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1-Tube inlet	
Dist. from "FROM" Node / Offset dist	75	cm
Nozzle Diameter	400	mm
Nozzle Schedule	160	
Nozzle Class	900	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	466.96	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2-Tube Outlet	
Dist. from "FROM" Node / Offset dist	75	cm
Nozzle Diameter	400	mm
Nozzle Schedule	160	
Nozzle Class	900	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	466.96	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Weight	
Detail ID	baffle	
Dist. from "FROM" Node / Offset dist	75	cm
Miscellaneous Weight	150	kgf
Offset from Element Centerline	0	mm

Element#	4/5	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	650	cm
Inside Diameter	1400	mm
Element Thickness	88	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	88	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	157.49	kgf/cm ²
Design Temperature Internal Pressure	93	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	93	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	162.5	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	1200	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	600	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	487.5	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	1200	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	600	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	140	cm
Liquid Density	992.24	kg/m ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	650	cm
Thickness of Insulation	25	mm
Density	0.00015	kg/cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	70	cm
Nozzle Diameter	400	mm
Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	270.0	



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 1040.5 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID N4-Shell Outlet
 Dist. from "FROM" Node / Offset dist 500 cm
 Nozzle Diameter 400 mm
 Nozzle Schedule 160
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element# 5/5
 Element From Node 50
 Element To Node 60
 Element Type Elliptical
 Description Dish End
 Distance "FROM" to "TO" 5.08 cm
 Inside Diameter 1400 mm
 Element Thickness 76 mm
 Internal Corrosion Allowance 3.175 mm
 Nominal Thickness 90 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 157.49 kgf/cm²
 Design Temperature Internal Pressure 93 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 93 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 138 cm
 Liquid Density 992.24 kg/m³

Element From Node 50
 Detail Type Insulation
 Detail ID Insulation
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 39.58 cm
 Thickness of Insulation 25 mm
 Density 0.00015 kg/cm³



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
LHS Ch Cover		31.8175	...	31.5	...
LHS Ch flg		71.3175	...	39.5	...
ch shell		221.317	...	150	...
Main Shell		903.318	...	650	...
Dish End		908.398	...	5.08	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: New Flange :

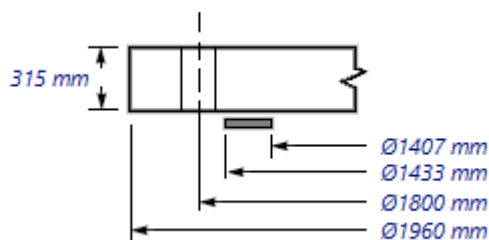
LHS Ch Cover

Flange Type	TEMA Channel Cover
Design Pressure	P 121.34 kgf/cm ²
Design Temperature	66 °C
Internal Corrosion Allowance	ci 3.1750 mm
External Corrosion Allowance	ce 0.0000 mm
Use Corrosion Allowance in Thickness Calcs.	Yes
Flange Outside Diameter	A 1960.000 mm
Flange Thickness	t 315.0000 mm
Flange Material	SA-516 70 +IT
Flange Material UNS number	K02700
Flange Allowable Stress At Temperature	Sfo 1407.13 kgf/cm ²
Flange Allowable Stress At Ambient	Sfa 1407.13 kgf/cm ²
Bolt Material	SA-193 B7
Bolt Allowable Stress At Temperature	Sb 1621.26 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa 1621.26 kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L 5654.867 mm
Youngs Modulus for Blind Flange Matl.	Sy 2027006.25 kgf/cm ²
Allowed Channel Cover Deflection	dfAllow 1.7500 mm
Diameter of Bolt Circle	C 1800.000 mm
Nominal Bolt Diameter	a 82.5500 mm
Type of Threads	TEMA Thread Series
Number of Bolts	32
Flange Face Outside Diameter	Fod 1444.000 mm
Flange Face Inside Diameter	Fid 1400.000 mm
Flange Facing Sketch	Sketch 1a
Gasket Outside Diameter	Go 1433.000 mm
Gasket Inside Diameter	Gi 1407.000 mm
Gasket Factor	m 3.0000
Gasket Design Seating Stress	y 703.07 kgf/cm ²
Gasket Thickness	tg 3.1750 mm
Raised face depth	Fd 0.0000 mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb*\sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 * b$	1420.151 mm



Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1420^2 \cdot 121$$

$$= 1922017.375 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1420 \cdot 3 \cdot 121$$

$$= 208678.328 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1922017 + 208678 + 0, 0)$$

$$= 2130695.750 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 6.42 \cdot \pi \cdot 1420 + 0 \cdot 0 \cdot 0$$

$$= 201523.453 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((2130695 + 0 + 4 \cdot 0/1420)/1621, 201523/1621)$$

$$= 1314.221 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1314.221	1545.700	
Radial Distance between Bolts and Edge:	76.200	80.000	
Circ. Spacing between the Bolts:	171.450	176.431	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1546 \cdot 1621 / (703 \cdot \pi (1433 + 1407))$$

$$= 39.950 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1546$$

$$= 2505983.50 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2130695.75 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1800 - 1420) / 2$$

$$= 189.9245 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 2130695 \cdot 190 \cdot 1 / (312^2 \cdot 1420) + 0.3 \cdot 1 \cdot 121 \cdot 1420^2 / 312^2$$

$$= 1311.84 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2505983 \cdot 190 \cdot 1 / (312^2 \cdot 1420)$$

$$= 654.87 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:



= Wm1 / Ab
 = 2130695/1546
 = 1378.47 kgf/cm²

Bolt Stress, Seating [BSa]:
 = (Wm2 / Ab)
 = (201523/1546)
 = 130.38 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1311.84	1407.13	654.87 1407.13
Bolting	1378.47	1621.26	130.38 1621.26

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.8992 mm
 Allowed deflection at center of Cover 1.7500 mm

Reqd. Blind Flange Thickness at Center 304.256 mm
 Reqd. Blind Flange Thickness at Gasket 215.902 mm
 Min. Required Flange Thickness per TEMA 249.758 mm

Estimated M.A.W.P. (Operating) 130 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 7366.1 kg
 Estimated Unfinished Weight of Forging at given Thk 7366.1 kg

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1285.53	1407.13	641.74 1407.13
Bolting	1378.47	1621.26	130.38 1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 301.092 mm

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Flange Input Data Values

Description: FLANGE :

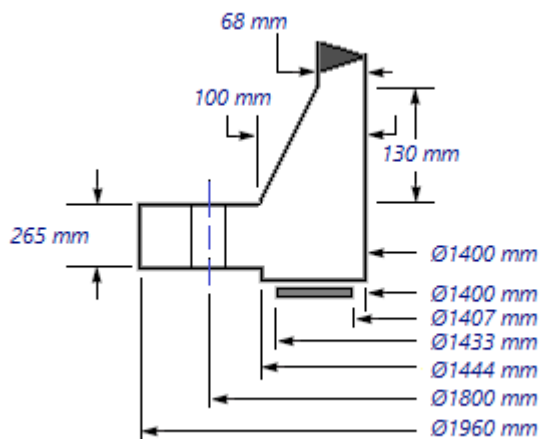
LHS Ch flg

Flange Type	Integral Weld Neck		
Design Pressure	P	121.28	kgf/cm ²
Design Temperature		66	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1400.000	mm
Flange Outside Diameter	A	1960.000	mm
Flange Thickness	t	265.0000	mm
Thickness of Hub at Small End	go	68.0000	mm
Thickness of Hub at Large End	gl	100.0000	mm
Length of Hub	h	130.0000	mm
Flange Material		SA-266 2 +IT	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1589.87	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1682.44	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1800.000	mm
Nominal Bolt Diameter	a	82.5500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1444.000	mm
Flange Face Inside Diameter	Fid	1400.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1433.000	mm
Gasket Inside Diameter	Gi	1407.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P(R + Ca)/(S - 0.5 \cdot P)) + (Ca + Ca_{ext}) \\ &= (121(1400 + 3.17)/(1407 - 0.5 \cdot 121)) + (3.17) \\ &= 66.5131 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= S \cdot \log((2 \cdot T + R)/R) \\ &= 1407 \cdot \log((2 \cdot 64.8 + 703)/703) \\ &= 124.086 \text{ kgf/cm}^2 \end{aligned}$$

Small End Hub MAPnc:

$$\begin{aligned} &= S_a \cdot \log((2.0 \cdot T + R)/R) \\ &= 1407 \cdot \log((2.0 \cdot 68 + 700)/700) \\ &= 130.454 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange Thickness, $t_c = T - c_i$	261.825 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1406.350 mm
Corroded Large Hub, $g_{lcor} = g_l - c_i$	96.825 mm
Corroded Small Hub, $g_{s cor} = g_s - c_i$	64.825 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g_{lcor}$	100.000 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1420.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 1420^2 \cdot 121 \\ &= 1921152.875 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 6.42 \cdot \pi \cdot 1420 \cdot 3 \cdot 121 \\ &= 208584.484 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot B_{cor}^2 \cdot P \\ &= \pi/4 \cdot 1406^2 \cdot 121 \\ &= 1883995.250 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - H_d \\ &= 1921152 - 1883995 \\ &= 37157.578 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(W_o, 1 + W_o, 2 + H'p, 0) \\ &= \max(1921152 + 208584 + 0, 0) \\ &= 2129737.250 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\ &= 703 \cdot 6.42 \cdot \pi \cdot 1420 + 0 \cdot 0 \cdot 0 \\ &= 201523.453 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a) \\ &= \max((2129737 + 0 + 4 \cdot 0/1420)/1621, 201523/1621) \\ &= 1313.630 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 82.5 + 6 \cdot 262/(3 + 0.5) \\ &= 613.943 \text{ mm} \end{aligned}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1800 \cdot \sin(\pi / 32)$$

$$= 176.431 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{176 / (2 \cdot 82.5 + 262)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1313.630	1545.700	
Radial Distance between Hub and Bolts:	95.250	100.000	
Radial Distance between Bolts and Edge:	76.200	80.000	
Circ. Spacing between the Bolts:	171.450	176.431	613.943

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1546 \cdot 1621 / (703 \cdot \pi(1433 + 1407))$$

$$= 39.950 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1546$$

$$= 2505983.50 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 2129737 - 1921152$$

$$= 208584.52 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1800 - 1420) / 2$$

$$= 189.9245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1800 - 1406) / 2 + 190)$$

$$= 193.3748 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1800 - 1406 - 96.8) / 2$$

$$= 148.4125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1883995.	148.4125	1.0000	279609.
Face Pressure, Mt	37158.	193.3748	1.0000	7185.
Gasket Load, Mg	208585.	189.9245	1.0000	39615.
Gasket Seating, Matm	2505984.	189.9245	1.0000	475948.

Total Moment for Operation, Mo 326409. kgf-m.
 Total Moment for Gasket seating, Mg 475948. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 301.938 mm
 Hub Ratio, h/h0 = HL / H0 0.431



Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.494

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.858
Factor V per 4.16.5 0.332
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 1960/1406
= 1.394

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
= $(0.66845 + (5.7169(1.39^2 \cdot \log_{10}(1.39))) / (1.39^2 - 1)) / (1.39 - 1)$
= 6.013

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.39^2(1 + 8.55246 \cdot \log_{10}(1.39)) - 1) / ((1.0472 + 1.9448 \cdot 1.39^2) \cdot (1.39 - 1))$
= 1.757

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.39^2(1 + 8.55246 \cdot \log_{10}(1.39)) - 1) / (1.36136(1.39^2 - 1) \cdot (1.39 - 1))$
= 6.608

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.39^2 + 1) / (1.39^2 - 1)$
= 3.122

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
= $6.61 \cdot 64.8^2 \cdot 302 / 0.33$
= 25270066.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.86/302
= 0.003 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
= $(262 \cdot 0.0028 + 1) / 1.76 + 262^3/25270066$
= 1.703

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
= $(1 \cdot 326409/1406) / (1.7 \cdot 96.8^2)$
= 1453.72 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
= $(1 \cdot 475948/1406) / (1.7 \cdot 96.8^2)$
= 2119.71 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(1.99 \cdot 326409/1406) / (1.7 \cdot 262^2)$
= 396.08 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(1.99 \cdot 475948/1406) / (1.7 \cdot 262^2)$
= 577.53 kgf/cm²

Tangential Flange Stress, Operating [STo]:



$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (6.01 \cdot 326409 / (262^2 \cdot 1406)) - 3.12 \cdot 396$$

$$= 799.13 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (6.01 \cdot 475948 / (262^2 \cdot 1406)) - 3.12 \cdot 578$$

$$= 1165.24 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1454 + \max(396, 799)) / 2$$

$$= 1126.42 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STa)) / 2$$

$$= (2120 + \max(578, 1165)) / 2$$

$$= 1642.48 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 2129737 / 1546$$

$$= 1377.85 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (201523 / 1546)$$

$$= 130.38 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1453.72	2384.80	2119.71	2523.66
Radial Flange	396.08	1589.87	577.53	1682.44
Tangential Flange	799.13	1589.87	1165.24	1682.44
Maximum Average	1126.42	1589.87	1642.48	1682.44
Bolting	1377.85	1621.26	130.38	1621.26

Minimum Required Flange Thickness 259.461 mm

Estimated M.A.W.P. (Operating) 124 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 3429.8 kg
 Estimated Unfinished Weight of Forging at given Thk 4524.1 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 475948 / 1 \cdot 100000 \cdot 0.33 / (1.65 \cdot 2067025 \cdot 64.8^2 \cdot 302 \cdot 0.3)$$

$$= 0.634 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 326409 / 1 \cdot 100000 \cdot 0.33 / (1.65 \cdot 2041067 \cdot 64.8^2 \cdot 302 \cdot 0.3)$$

$$= 0.440 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 475948 / 1 \cdot 100000 \cdot 0.33 / (1.7 \cdot 2067025 \cdot 64.8^2 \cdot 302 \cdot 0.3)$$

$$= 0.614 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 326409 / 1 \cdot 100000 \cdot 0.33 / (1.7 \cdot 2041067 \cdot 64.8^2 \cdot 302 \cdot 0.3)$$

$$= 0.426 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1411.22	2523.66	2033.99	2523.66
Radial Flange	399.12	1682.44	575.25	1682.44
Tangential Flange	762.89	1682.44	1099.55	1682.44
Maximum Average	1087.05	1682.44	1566.77	1682.44
Bolting	1377.85	1621.26	130.38	1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 247.777 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
LHS Ch Cover		121.34	323	3.175	1960	1407.1	1
LHS Ch flg		121.28	273	3.175	1400	1589.9	2
ch shell		121.28	68	3.175	1400	1407.1	3
Main Shell		157.63	88	3.175	1400	1407.1	4
Dish End		157.63	90	3.175	1400	1407.1	5

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
LHS Ch Cover		121.145	130	133	315	304.256	1
LHS Ch flg		121.145	124	130	265	259.461	2
ch shell		121.145	124	130	68	66.4723	3
Main Shell		157.489	160	167	88	86.5267	4
Dish End		157.489	158	165	76	75.9636	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Note:

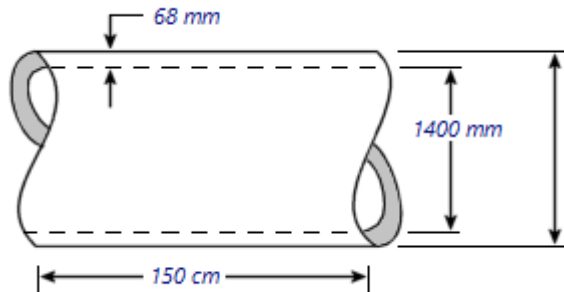
The MAWPs and MAPs for the Exchanger elements all appear to be zero or some could not be calculated. Please fill these values in on the main tubesheet tab.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: ch shell

Cylindrical Shell Design Information:					
Design Pressure	P	121.284 kgf/cm ²	Int. Design Temperature	66 °C	
Material		SA-516 70	Ext. Design Temperature	66 °C	
Length		150.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	1400.000 mm	Actual Stress	1315.6 kgf/cm ²	
Spec. Min. Thk t _e , or sa		68.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		72382.3 cm ²	SG of Contents	0.993	
Element Volume		2309070.8 cm ³	Weight of Contents	2291.2 kgf	
Empty Weight		3645.8 kgf	Operating Weight	5937.0 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm ²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1406 \cdot (\exp(121/(1407 \cdot 1)) - 1) + 3.17 + 0$
 $= 63.3 + 3.17 + 0 = 66.5 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 64.8 + 1406) / 1406) - 0.14$
 $= 124 - 0.14 = 124 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 68 + 1400) / 1400)$
 $= 130 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 121 \cdot 1406 / (1 (1536 - 1406))$
 $= 1315.6053 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 121 / (1 \cdot \ln((2 \cdot 64.8 + 1406) / 1406))$$
$$= 1375.3556 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.632 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

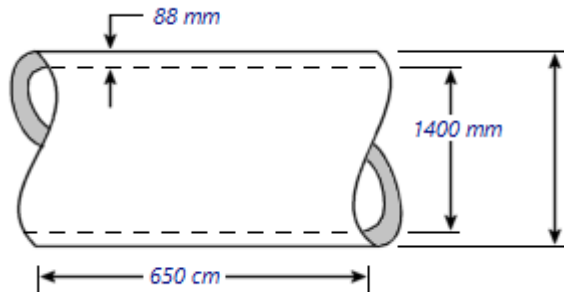
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:				
Design Pressure	P	157.628 kgf/cm²	Int. Design Temperature	93 °C
Material		SA-516 70	Ext. Design Temperature	93 °C
Length		650.000 cm	External Pressure Chart	CS-2
			UNS Number	K02700
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope) S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1407.1 kgf/cm²
Inside Diameter	Di	1400.000 mm	Actual Stress	1306.7 kgf/cm²
Spec. Min. Thk t,e, or sa		88.0000 mm	Spec. Nominal Thk.	88.000 mm
Surface Area		321824.8 cm²	SG of Contents	0.993
Element Volume		5522123.5 cm³	Weight of Contents	5479.4 kgf
Empty Weight		20723.9 kgf	Operating Weight	26203.2 kgf
			Comp. All. Ope. Stress	1251.5 kgf/cm²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1406 \cdot (\exp(158/(1407 \cdot 1)) - 1) + 3.17 + 0$
 $= 83.4 + 3.17 + 0 = 86.5 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 84.8 + 1406) / 1406) - 0.14$
 $= 160 - 0.14 = 160 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 88 + 1400) / 1400)$
 $= 167 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 158 \cdot 1406 / (1 (1576 - 1406))$
 $= 1306.6901 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 158 / (1 \cdot \ln(2 \cdot 84.8 + 1406) / 1406)$$
$$= 1384.0082 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.914 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

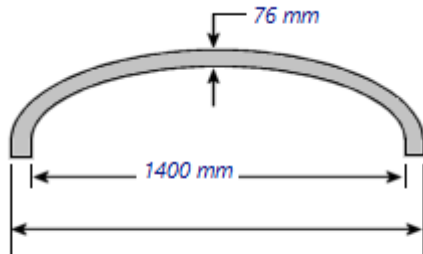
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	157.626 kgf/cm²	Int. Design Temperature	93 °C	
Material		SA-516 70	Ext. Design Temperature	93 °C	
Length		5.080 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1400.000 mm	Actual Stress	1406.4 kgf/cm²	
Spec. Min. Thk t _e , or s _a		76.0000 mm	Spec. Nominal Thk.	90.000 mm	
Surface Area		29582.2 cm²	SG of Contents	0.993	
Element Volume		437389.3 cm³	Weight of Contents	433.7 kgf	
Empty Weight		2019.3 kgf	Operating Weight	2453.0 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	
			Aspect Ratio	2.00	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 17.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.750 mm

Material treatment options: +IT

Warning: L/t out of range - see 4.3.6.1(b)!

Warning: 20 ≤ L/t ≤ 2000; Actual = 17.3 ; This requirement does not apply to heads designed as dished bolted covers per Part 4.7.

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

tr = 72.8 mm

t = tr + ci + co
 = 72.8 + 3.17 + 0
 = 76 mm - see below for the derivation

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

h = D/(2 • Aspect Ratio)
 = 1400/(2 • 2)
 = 350.000 mm

k = D/(2 • h)



$$= 1400 / (2 \cdot 350)$$

$$= 2.000$$

$$r = D (0.5/k - 0.08) + ca$$

$$= 1400 (0.5/2 - 0.08) + 3.17$$

$$= 241.175 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 1400 (0.44 \cdot 2 + 0.02) + 3.17$$

$$= 1263.175 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 72.8 \text{ mm}$, $P = 158 \text{ kgf/cm}^2$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r) / (L - r)]$$

$$= \arccos[(0.5 \cdot 1406 - 241) / (1263 - 241)]$$

$$= 1.1 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1263 \cdot 72.8) / 241}$$

$$= 1.26 \text{ radians - Equation 4.3.9}$$

$\phi Th > \beta Th$, Thus:

$$Rth = 0.5 \cdot D = 0.5 \cdot 1406 = 703 \text{ mm - Equation 4.3.11}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$$r/D > 0.08, \text{ Thus:}$$

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(241/1406) + 0.605$$

$$= 0.72 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(241/1406)$$

$$= 1.01, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.72 \cdot 2024848 \cdot 72.8^2 / (1.01 \cdot 703 (0.5 \cdot 703 - 241))$$

$$= 98602 \text{ kgf/cm}^2, \text{ Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 2451 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 2451 \cdot 72.8 / (1.01 \cdot 703 (0.5 \cdot 703 / 241 - 1))$$

$$= 546 \text{ kgf/cm}^2, \text{ Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 98602/546 = 180$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 180 - 0.20354 \cdot 180^2 + 0.019274 \cdot 180^3) /$$

$$(1 + 0.19014 \cdot 180 - 0.089534 \cdot 180^2 + 0.0093965 \cdot 180^3)) \cdot 546$$

$$= 1115 \text{ kgf/cm}^2, \text{ Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 1115/1.5 = 743 \text{ kgf/cm}^2 \text{ - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1407 \cdot 1 / (1263/72.8 + 0.5)$$

$$= 158 \text{ kgf/cm}^2, \text{ Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(743, 158)$$



= 158 kgf/cm², Equation 4.3.23: (same as original design pressure)

Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1407 \cdot 1 / (1263 / 72.8 + 0.5) \\ &= 157.703 \text{ kgf/cm}^2 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 158 - 0.14 = 158 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 165 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 1406.4 \text{ kgf/cm}^2$$

Required Thickness of Straight Flange = 86.526 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	23.852 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	32.660 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 29582 / 3.14159) \\ &= 1940.752 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 17.500 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.750 mm

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{ShellSide MAWP(St/S)}$$

$$= 1.25 \cdot 158 (1)$$

$$= 196.957 \text{ kgf/cm}^2$$

Hydrotest Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Channel Side MAWP(St/S)}$$

$$= 1.25 \cdot 122 (1)$$

$$= 152.753 \text{ kgf/cm}^2$$

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
ch shell	1649.2	2386.0	0.691	152.89
Main Shell	1664.4	2386.0	0.698	197.10
Dish End	1683.1	2386.0	0.705	197.10

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	1203.20	1203.20	1.000
N1-Tube inlet	Pad	1407.13	1407.13	1.000
N2-Tube Outlet	Nozzle	1203.20	1203.20	1.000
N2-Tube Outlet	Pad	1407.13	1407.13	1.000
N3-Shell Inlet	Nozzle	1407.13	1407.13	1.000
N4-Shell Outlet	Nozzle	1407.14	1407.14	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Ch Cover	1407.13	1407.13	1.000
LHS Ch flg	1682.44	1589.87	1.058
ch shell	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
Dish End	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	688.55	2202.47	0.313
N2-Tube Outlet	688.55	2202.47	0.313
N3-Shell Inlet	887.62	2294.24	0.387
N4-Shell Outlet	887.62	2294.24	0.387



Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		196.96 kgf/cm ²
Shell Thickness	ts	88.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1400.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1784.41 kgf/cm ²
Shell Allowable Stress at Ambient		1784.41 kgf/cm ²

Channel Data:

Channel Description: ch shell

Channel Type:		Cylinder
Channel Test Pressure		152.75 kgf/cm ²
Channel Thickness	tc	68.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1400.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1784.41 kgf/cm ²
Channel Allowable Stress at Ambient		1784.41 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	2298
Tube Wall Thickness	et	3.0480 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6096.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5776.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Is this a Welded Tube		No
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1413.25 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0480 mm
Groove Weld Leg	ag	3.0480 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	681.025 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-266 2

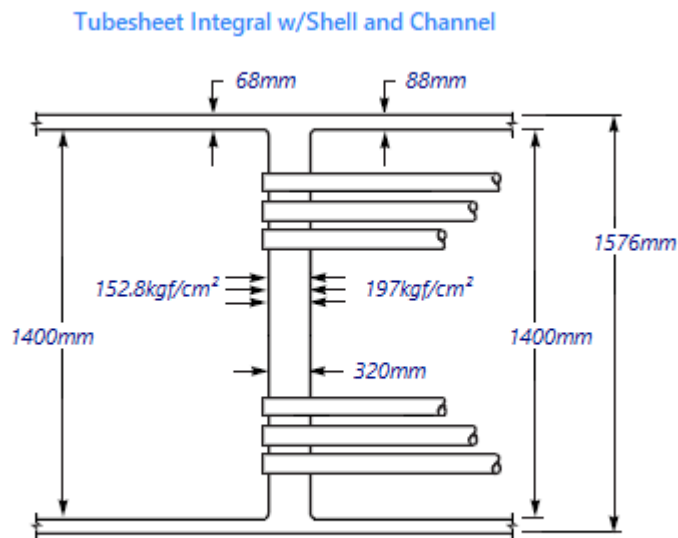
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	320.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1576.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20530E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20530E+07 kgf/cm ²

Iteration

	Fic	H	D1	D2	D3	D4	Modulus	Diff
1	1652	-1.181	0.753E-05	0.375E-05	0.148E-04	-0.319E-05	0.428E+05	0.161E-01
2	826	-4.012	0.355E-06	0.354E-06	0.393E-05	-0.392E-05	0.833E+06	0.215E-01
3	1239	-2.597	0.212E-05	0.206E-05	0.853E-05	-0.816E-05	0.199E+06	0.163E-01
4	1445	-1.889	0.418E-05	0.366E-05	0.115E-04	-0.924E-05	0.949E+05	0.729E-02
5	1548	-1.535	0.567E-05	0.422E-05	0.131E-04	-0.787E-05	0.642E+05	0.159E-02
6	1497	-1.712	0.488E-05	0.400E-05	0.123E-04	-0.884E-05	0.782E+05	0.339E-02
7	1523	-1.623	0.526E-05	0.413E-05	0.127E-04	-0.844E-05	0.709E+05	0.105E-02
8	1536	-1.579	0.546E-05	0.418E-05	0.129E-04	-0.818E-05	0.675E+05	0.227E-03
9	1529	-1.601	0.536E-05	0.416E-05	0.128E-04	-0.832E-05	0.692E+05	0.423E-03
10	1532	-1.590	0.541E-05	0.417E-05	0.128E-04	-0.825E-05	0.683E+05	0.101E-03
11	1534	-1.585	0.543E-05	0.418E-05	0.128E-04	-0.821E-05	0.679E+05	0.625E-04
12	1533	-1.587	0.542E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.193E-04
13	1534	-1.586	0.543E-05	0.418E-05	0.128E-04	-0.822E-05	0.680E+05	0.215E-04
14	1533	-1.587	0.543E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.110E-05
15	1533	-1.587	0.542E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.911E-05
16	1533	-1.587	0.542E-05	0.418E-05	0.128E-04	-0.823E-05	0.681E+05	0.400E-05

Factor of Safety [FS]:
 Since Fic [1534] >= Fy[1413], FS = 1.667

Allowable Circumferential Compressive Stress [Fha]:
 = Fic/FS
 = 1534/1.67
 = 920.081 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:
 = 2 • Fha(t / Do)
 = 2 • 920 (3.05/19)
 = 294.426 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.2504 mm
Allowable Internal Pressure at Corroded thickness	363.36 kgf/cm ²
Required Internal Design Pressure	152.75 kgf/cm ²
Allowable External Pressure at Corroded thickness	294.43 kgf/cm ²
Required External Design Pressure	196.96 kgf/cm ²
Required Thickness due to Shell Side pressure	2.2504 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 = 2 • ro + dt
 = 2 • 681 + 19
 = 1381.100 mm

Basic ligament efficiency for shear [μ]:
 = (p - dt) / p
 = (23.8 - 19)/23.8
 = 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:
 = Ds / Do (Configurations a, b, c)
 = 1400/1381
 = 1.0137

Ratio [pc]:
 = Dc / Do (Configurations a, e, f)



$$= 1400/1381$$

$$= 1.0137$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/320$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t)$$

$$= \max(19 - 2 \cdot 3.05 (2067025/2052964) \cdot (942/1407) \cdot (0); 19 - 2 \cdot 3.05)$$

$$= 19.0500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 23.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1381 \cdot 23.8) / (\pi \cdot 1381^2)) }$$

$$= 23.8120 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (23.8 - 19) / 23.8$$

$$= 0.2000$$

$$h/p = 13.438601 ; \mu^* = 0.199983 ; E^* = 318885 \text{ kgf/cm}^2$$

$$E^*/E = 0.155329 ; \nu^* = 0.458596$$

Note: As h/p (13.439) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [βs]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1400 + 88) 88 }$$

$$= 0.0050 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.005 \cdot 2067025 \cdot 88^3 / (6 (1 - 0.3^2))$$

$$= 1296027392.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (1400 + 88) \cdot 1296027392 / 320^3 \cdot (1 + 320 \cdot 0.005 + (320 \cdot 0.005)^2 / 2)$$

$$= 1377029.5 \text{ kgf/cm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 1400^2 / (4 \cdot 2067025 \cdot 88) \cdot (1 - 1400 \cdot 0.3 / (2 (1400 + 88)))$$

$$= 0.23136E-02 \text{ mm/kgf/cm}^2$$

Parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 1296027392 \cdot 0.005 \cdot 0.002313638 (1 + 320 \cdot 0.005)$$

$$= 39815.9336 \text{ mm}^2$$

Channel coefficient [βc]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1400 + 68) 68 }$$

$$= 0.0058 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6 (1 - \nu_s^2))$$

$$= 0.0058 \cdot 2067025 \cdot 68^3 / (6 (1 - 0.3^2))$$

$$= 684885952.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$



Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (C_m \theta + h \cdot \beta_c \cdot (C_m \Delta + C_q \theta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot C_q \Delta)$$

$$= 6(1400 + 68 / \cos(0)) \cdot 684885952 / 320^3 \cdot (1 + 320 \cdot 0.0058 \cdot (1 + 1) / 2 + (320 \cdot 0.0058)^2 / 2 \cdot 1)$$

$$= 835072.0 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1400^2 / (4 \cdot 2067025 \cdot 68) \cdot (1 - 1400 \cdot 0.3 / (2(1400 + 68)))$$

$$= 0.29874E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (C_m \theta + \beta_c + C_q \Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m \theta + h \cdot C_q \theta + h \cdot \beta_c / 2)]$$

$$= 1.01 \cdot 684885952 [0.002987417 (1 + 0.0058 \cdot 1 \cdot 320) - 0.000 (1/320 + 1 \cdot 0.0058/2)]$$

$$= 33903.8125 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1576 / 1381$$

$$= 1.1411$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 318886 \cdot (1377029 + 835072 + 2052964 \cdot \ln(1.14))$$

$$= 4.2158$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1381^2 / 16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 197 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 153]$$

$$= 146216.4531 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 146216 + 33904 \cdot 153 - 39816 \cdot 197$$

$$= -2516894.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - D_o^2 / 32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= ((-2516894) - 1381^2 / 32 \cdot 4.22 (197 - 153)) / (1 + 4.22)$$

$$= -2612256.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= -2612256 + 1381^2 / 64 \cdot (3 + 0.46) \cdot (197 - 153)$$

$$= 1944230.2500 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p |; | M_o |)$$

$$= \max(| -2612256 |; | 1944230 |)$$

$$= 2612256.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$



$$= 6 \cdot 2612256 / (0.2 (320 - 6)^2) \\ = 794.9031 \text{ kgf/cm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y \\ = 2 \cdot 0.95 \cdot 2549 \\ = 4843.40 \text{ kgf/cm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ = 6 \cdot 2597395 / (0.2 (133 - 6)^2) \\ = 4828.4810 \text{ kgf/cm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc} \\ = 133 + 0 + 0 \\ = 133.0406 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS}) \\ = \max(133; 39.4) \\ = 133.0406 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t| \\ = |197 - 153| \\ = 44.204 \text{ kgf/cm}^2$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ = 4 \cdot 0.2 \cdot 320 / 1381 \cdot 0.8 \cdot 0.95 \cdot 2549 \\ = 359.078 \text{ kgf/cm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ = 1 / (4 \cdot 0.2) \cdot (1381 / 1937) \cdot |197 - 153| + 0 \\ = 39.3932 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 88.000mm

Min. Shell len. adjacent to the tubesheet is: 631.797 mm

Note: For a given Channel thickness of: 68.000 mm

Min. Channel len. adjacent to the tubesheet is: 555.381mm

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t \\ = \pi (19 - 3.05) 3.05 \\ = 1.532 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S \\ = \pi \cdot 3.05 (19 - 3.05) 1343 \\ = 2057.225 \text{ kgf}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ = 0.85 \cdot \pi \cdot 3.05 (19 + 0.67 \cdot 3.05) 1343 \\ = 2057.2249 \text{ kgf}$$

Weld Strength Factor [f_w]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

= 1343/1343
 = 1.000

Design Strength Ratio [fd]:
 = 1 for full strength welds

Fillet Weld Strength [Ff]:
 = $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 = $0.55 \cdot \pi \cdot 3.05 (19 + 0.67 \cdot 3.05) 1343$
 = 1491.3901 kgf

Minimum Length of Weld [ar]:
 = $2 \cdot \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd) - 0.75 \cdot do}$
 = $2 \cdot \sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1) - 0.75 \cdot 19}$
 = 3.445 mm

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm
 Minimum Required Groove Weld Leg [agr]: 1.8288 mm

Tube-Tubesheet Jt allowable, 2057 is >= tube strength 2057 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	794.9 <=	4843.4 kgf/cm ²	Ok
Tubesheet shear stress	238.5 <=	1937.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	133.041	320.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.048	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.048	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses			Case Type	Pass/Fail
		Bend	Allwd	Shear	Allwd		
D1uc 235.410	...	2512	4843	824	1937	Ps+Pt D1	Ok
D2uc 266.612	...	3307	4843	1063	1937	Ps+Pt D2	Ok
D3uc 133.041	...	795	4843	238	1937	Ps+Pt D3	Ok
Max: 266.6119	... mm		0.683		0.549	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.							
Case#	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	3778	...	3778	Ok
D2uc	...	...	...	3778	...	3778	Ok
D3uc	...	...	...	3778	...	3778	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	266.612	320.000	Ok
Tube thickness :	2.250	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.048	Ok
Tube-tubesheet groove weld leg :	1.829	3.048	Ok

Given Shell Thickness: 88.0000 mm

Given Channel Thickness: 68.0000 mm

Min Shell length of thk, (88.000) adj. to tubesheet: 631.797 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 555.381 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	311.825	No Calc	No Calc
20	30	No Calc	...	261.825	No Calc	No Calc
30	Tsht	189.497	1536	64.825	No Calc	No Calc
Tsht	40	No Calc	...	64.825	No Calc	No Calc
40	50	666.75	1576	84.825	No Calc	No Calc
50	60	No Calc	1552	72.825	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	315	215.902	...	No Calc
20	30	265	259.461	...	No Calc
30	Tsht	68	No Calc	...	109
Tsht	40	68	No Calc	...	No Calc
40	50	88	No Calc	...	109
50	60	76	No Calc	...	125

Minimum

109

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	189.497	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	666.75	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	7553.13	...	7553.13	...	755.313
20	30	3521.39	624191	3477.67	626011	352.139
30	40	3645.85	2309070	3483.14	2330064	364.585
40	50	20723.9	5522123	20018.8	5613098	2072.39
50	60	2019.31	437389	1948.07	443010	201.931
Total		37463	8892775.00	36480	9012184.00	3746



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
10	Liqd	...	15.75	0.5	...	Liquid: 10
20	Liqd	619.357	19.75	...	...	Liquid: 20
30	Liqd	2291.19	75	...	...	Liquid: 30
30	Nozl	513.656	75	86.2738	...	N1-Tube inlet
30	Nozl	513.656	75	-86.2738	...	N2-Tube Outlet
30	Wght	149.997	75	...	...	baffle
40	Sadl	389.16	162.5	89.8	...	Left Saddle
40	Sadl	389.16	487.5	89.8	...	New Saddle
40	Liqd	5479.35	325	...	...	Liquid: 40
40	Insl	122.74	325	...	...	Hot Insulation
40	Nozl	1144.6	70	-86.2738	...	N3-Shell Inlet
40	Nozl	1144.59	500	86.2738	...	N4-Shell Outlet
50	Liqd	433.738	16.7467	1	...	Liquid: 50
50	Insl	13.7513	19.79	...	...	Insulation
30	FTsh	3535.03	166	...	...	
30	Tube	20866.2	454.8	...	...	Tubes

Total Weight of Each Detail Type:

Saddles	778.3
Liquid	8823.6
Insulation	136.5
Nozzles	3316.5
Weights	150.0
Exchanger Components	24401.2
Liquid in Tubes	2153.4

Sum of the Detail Weights 39759.6 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	41209.9	41209.9	41209.9	41209.9	41209.9	41209.9
Saddles	778.3	778.3	778.3	778.3	778.3	778.3
Nozzles	3316.5	3316.5	3316.5	3316.5	3316.5	3316.5
Wld Weights	150.0	150.0	150.0	150.0	150.0	150.0
Exchanger	24401.2	24401.2	24401.2	24401.2	24401.2	24401.2
Insulation	...	...	136.5	136.5	136.5	136.5
Ope. Liquid	...	...	...	...	...	8823.6
Tube Ope Lqd	...	...	...	...	...	2153.4
Test Liquid	...	8888.8	...	...	...	...
Tube Tst Lqd	...	2169.3	...	...	...	...



Totals	69856.0	80914.0	69992.5	69992.5	69992.5	80969.5
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Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	69856.0 kg
Shop Test Wt.	- Fabricated weight + water (full)	80914.0 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	69992.5 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	69992.5 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	69992.5 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	80969.5 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	79888.5 kg
Field Test Wt.	- Empty weight + test medium (full)	72161.7 kg

Exchanger Tube Data

Volume of Exchanger tubes :	2170248.5 cm ³
Weight of Ope Liq in tubes :	2153.4 kg
Weight of Water in tubes :	2169.3 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	50060.6	...
20	30	38122.6	17724.9
30	40	72382.3	65973.5
40	50	321825	285885
50	60	29582.2	23480.4

Total	511972.5	393063.7 cm ²
Total	551.1	423.1 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-Tube inlet	149.33	156.12	66	900	GR 1.1	38.567	49.330	49.330	49.330
N2-Tube Outlet	149.33	156.12	66	900	GR 1.1	38.567	49.330	49.330	49.330
N3-Shell Inlet	240.21	260.14	93	1500	GR 1.1	47.255	40.207	40.207	40.207
N4-Shell Outle	240.21	260.14	93	1500	GR 1.1	47.255	40.207	40.207	40.207

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 240.207 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 260.136 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 149.330 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 156.122 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N1-Tube inlet	434.975	0.30	5766	4283	122.202	149.330	Pass
N2-Tube Outlet	434.975	0.30	5766	4283	122.202	149.330	Pass
N3-Shell Inlet	431.800	0.30	6920	5099	157.565	240.207	Pass
N4-Shell Outlet	431.800	0.30	6920	5099	157.565	240.207	Pass

Evaluating Flange on Nozzle [N1-Tube inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(5766) + 4 \cdot \text{abs}(4283) \cdot 434.975 \leq \pi \cdot 434.975^3 \cdot ((149.330 - 122.202) + 0.300 \cdot 149.330)$$

$$= 0.99713E+05 \leq 0.18596E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(149.330, 0.300 \cdot 149.330 + 149.330 - 99713.4 / (\pi \cdot 434.975^3))$$

$$= \min(149.330, 155.562)$$

$$= 149.330 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(5766) + 4 \cdot \text{abs}(4283) \cdot 434.975 \leq \pi \cdot 434.975^3 \cdot ((149.330 - 122.202) + 0.300 \cdot 149.330)$$

$$= 0.99713E+05 \leq 0.18596E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(149.330, 0.300 \cdot 149.330 + 149.330 - 99713.4 / (\pi \cdot 434.975^3))$$

$$= \min(149.330, 155.562)$$

$$= 149.330 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N3-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(6920) + 4 \cdot \text{abs}(5099) \cdot 431.800 \leq \pi \cdot 431.800^3 \cdot ((240.207 - 157.565) + 0.300 \cdot 240.207)$$

$$= 0.11952E+06 \leq 0.39129E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell Inlet [Pmax]:



$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(240.207, 0.300 \cdot 240.207 + 240.207 - 119520.4 / (\pi \cdot 431.800^3)) \\ &= \min(240.207, 265.015) \\ &= 240.207 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [N4-Shell Outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(6920) + 4 \cdot \text{abs}(5099) 431.800 \leq \pi \cdot 431.800^3 [(240.207 - 157.565) + .300 \cdot 240.207] \\ &= 0.11952\text{E}+06 \leq 0.39129\text{E}+06 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell Outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(240.207, 0.300 \cdot 240.207 + 240.207 - 119520.4 / (\pi \cdot 431.800^3)) \\ &= \min(240.207, 265.015) \\ &= 240.207 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	350	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 6.132

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.94 • 1 • 0.6 • 57.5²)

= 1867.1 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.74 • 0.82 • 1867

= 1138.0 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	4.70	1.0000	1.0000	1.0000	57.50	0.8226
20	4.70	1.0000	1.0000	1.0000	57.50	0.8226
30	4.70	1.0000	1.0000	1.0000	57.50	0.8226
40	4.70	1.0000	1.0000	1.0000	57.50	0.8226
50	4.70	1.0000	1.0000	1.0000	57.50	0.8226

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm²	Elevation cm	Pressure kgf/m²	Force kgf
baffle	0.00	470.00	190.47	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
10	20	470	235.2	7408.8	190.469	127.649
20	30	470	168	6636	190.469	114.335
30	40	470	184.32	27648	190.469	476.36
40	50	470	195.12	126828	190.469	2185.18
50	60	470	192.24	6651.1	190.469	114.595



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Soft Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.094 • 1.4) • 1
= 0.131

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 11301 • 0.13 • 1
= 1485.0 kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.13 • 79110 • 1
= 6930.0 kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height cm	Weight kgf	Ope Load kgf
10	20	98	11301.5	1484.99
20	30	70	11301.5	1484.99
30	40	70	11301.5	1484.99
40	Sad1	70	11301.5	1484.99
Sad1	50	70	11301.5	1484.99
40	50	70	11301.5	1484.99
50	60	70	11301.5	1484.99

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	578.318 cm
Center of Gravity of the Liquid	445.967 cm
Center of Gravity of the Insulation	613.055 cm
Center of Gravity of the Nozzles	416.892 cm
Center of Gravity of the Added Weights (Operating)	146.317 cm
Center of Gravity of the Added Weights (Empty)	146.317 cm
Center of Gravity of the Tubesheet(s)	237.317 cm
Center of Gravity of the Tubes	526.117 cm
Center of Gravity of Bare Shell New and Cold	391.947 cm
Center of Gravity of Bare Shell Corroded	388.813 cm
Vessel CG in the Operating Condition	430.926 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	427.296 cm
Vessel CG in the Test Condition	432.002 cm

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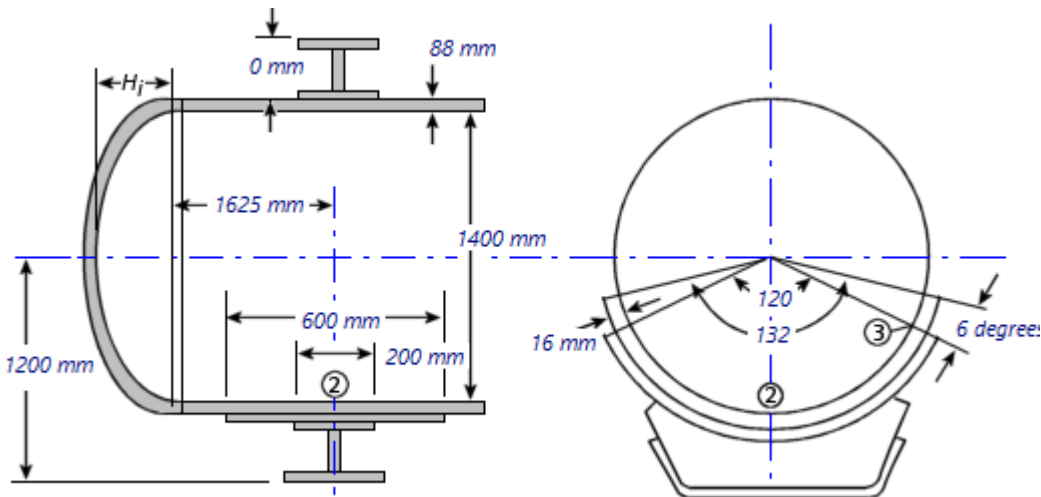


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	744.00	mm
Shell Thickness used in this Case	t	88.000	mm
Stiffened Vessel Length per 4.15.6	L	655.08	cm
Distance from Saddle to Vessel tangent	a1 or a	1625.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1630.99	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1589.87	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		2008.94	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	66898.44	kgf
Pressure used in Analysis	P	0.000 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-2.40	-2518.57
Long. Stress at Top of Midspan	2.40	1407.13
Long. Stress at Bottom of Midspan	2.40	1407.13
Long. Stress at Top of Saddles	147.97	1407.13
Long. Stress at Bottom of Saddles	-82.01	-2518.57
Long. Stress at Bottom of Saddles	82.01	1407.13
Tangential Shear in Shell	60.27	1125.70
Circ. Stress at Horn of Saddle	75.87	2110.69



Circ. Compressive Stress in Shell	8.16	1407.13
-----------------------------------	------	---------

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

= Bundle Pull Out Force • Impact Factor • B / Ls

= 2009 • 1.5 • 1200/3250

= 1112.6 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 65786 + max(1113, 0, 0, 0)

= 66898.4 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

67287.59 kgf

Transverse Shear Load

0.00 kgf

Longitudinal Shear Load

3013.41 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 1260.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066

K2 = 1.1707

K3 = 0.8799

K4 = 0.4011

K5 = 0.7603

K6 = 0.0529

K7 = 0.0529

K8 = 0.3405

K9 = 0.2711

K10 = 0.0581

K1* = 0.1923

K6p = 0.0434

K7p = 0.0434

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$ = $-66898 \cdot 162 [1 - (1 - 162/655 + (74.4^2 - 0^2)/(2 \cdot 162 \cdot 655))/(1 + (4 \cdot 0)/(3 \cdot 655))]$

= -24140.3 kgf-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$ = $66898 \cdot 655/4 ((1 + 2(74.4^2 - 0^2)/(655^2))/(1 + (4 \cdot 0)/(3 \cdot 655)) - 4 \cdot 162/655)$

= 3676.0 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$ = $0 \cdot 744/(2 \cdot 88) - 3676/(\pi \cdot 744^2 \cdot 88)$ = -2.40 kgf/cm²Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$ = $0 \cdot 744/(2 \cdot 88) + 3676/(\pi \cdot 744^2 \cdot 88)$ = 2.40 kgf/cm²Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:= $P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$ = $0 \cdot 744/(2 \cdot 88) - 24140/(0.11 \cdot \pi \cdot 744^2 \cdot 88)$ = 147.97 kgf/cm²Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:= $P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$ = $0 \cdot 744/(2 \cdot 88) + 24140/(0.19 \cdot \pi \cdot 744^2 \cdot 88)$ = -82.01 kgf/cm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm)/3)$



$$= 66898 (655 - 2 \cdot 162) / (655 + (4 \cdot 0/3))$$

$$= 33708.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 33709 / (744 \cdot 88)$$

$$= 60.27 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(744 \cdot 88)}$$

$$= 199.582 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(744 \cdot 88)}, 2 \cdot 162)$$

$$= 599.16 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1631/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 66898 \cdot 0.1 / (599(88 + 1 \cdot 16))$$

$$= -8.16 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -66898 / (4(88 + 1 \cdot 16)599) -$$

$$3 \cdot 0.053 \cdot 66898 / (2(88 + 1 \cdot 16)^2)$$

$$= -75.87 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(700 \cdot 88)} + 600)88^3) / 12$$

$$= 5606.134 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 5606 + 869(97 - 44)^2$$

$$= 30041.402 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	987.2	88.0	44.0	868.7	3822367.5	0.300E+05
Wearplate	600.0	16.0	96.0	96.0	921599.9	21.5
Web	10.0	362.0	285.0	36.2	1031700.1	0.167E+05
BasePlate	300.0	34.0	483.0	102.0	4926600.0	0.152E+06
Totals	...	...	...	1102.9	10702268.0	0.199E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10702268 / 1103$$



$$= 97.036 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 66898$$

$$= 13615.2822 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 58.1354 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1200 - 700 \cdot \sin(120/2) / 1.05$$

$$= 621.105 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8456.5244 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 41.2662 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(66898 + 389) \cdot 300) / (4 \cdot 1260 \cdot 1767)}$$

$$= 26.074 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (700 + 88 + 16)$$

$$= 1392.569 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1393 / (5 - 1)$$

$$= 348.142 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 348 \cdot 300 / 2$$

$$= 522.213 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 67288 / (1260 \cdot 300)$$

$$= 17.801 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 522 \cdot 17.8$$

$$= 9295.895 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 17.4$$

$$= 36.407 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 9296/36.4$$



= 255.332 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 200 - 10
 = 190.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/36.4
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 200/2
 = 100.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{667/36.4}$
 = 42.792 mm

Length of Outer Rib, step 1, [Lv]:
 = Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1200 - cos(120/2)(700 + 88 + 16) - 34
 = 764.000 mm

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120/2) (700 + 88 + 16)$
 = 1392.569 mm

Length of Outer Rib, step 3, [Bd]:
 = Baseplate length - clearance
 = 1260 - 50.8
 = 1209.200 mm

Length of Outer Rib, step 4, [Dd]:
 $= (Lw - Bd) / 2$
 $= (1393 - 1209) / 2$
 = 91.684 mm

Length of Outer Rib [L]:
 $= \sqrt{Lv^2 + Dd^2}$
 $= \sqrt{764^2 + 91.7^2}$
 = 769.482 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 2024850 / 2651}$
 = 122.785

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 769 / 42.8$
 = 17.982

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 3013 / (2 \cdot 1260) \cdot 348 \cdot 769 / 2$
 = 160.171 kgf-m.



Compressive Allowable, $KL/r < C_c$ ($18 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (18) / (8 \cdot 123) - (18^3) / (8 \cdot 123^3))$$

$$= 1524 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 255 / 1524 + (160 \cdot 100 / 6666666) / 1767$$

$$= 0.304$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	348.1	10.0	0.0	34.8	0.0	2.90
Totals	...	...	...	53.8	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 53.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1200 - \cos(804/2) (34 + 0 + 0) - 0$$

$$= 362.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (669 / 53.8) }$$

$$= 35.271 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 362 / 35.3$$

$$= 10.263$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 3013 / (2 \cdot 1260)$$

$$= 1.196 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.2 \cdot 348 \cdot 362$$

$$= 150.704 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 151 / 66948$$

$$= 225.104 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.3)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (10.3) / (8 \cdot 123) - (10.3^3) / (8 \cdot 123^3))$$

$$= 1556 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 303 / 1556 + (151 \cdot 100 / 669) / 1767$$

$$= 0.322$$



Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	5.4000 cm ²
Saddle Load Q0 (Weight)	Q0	67287.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3013.4 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	Fl	3013.4 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3013 / (5.4 \cdot 4)$$

$$= 139.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (5.4 \cdot 4)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3013 / (1052 \cdot 4)$$

$$= 0.7159 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1200 \cdot 0 + 0$$

$$= 0.00 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0 / 67288$$

$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.72, 0]$$

$$= 0.7159 \text{ cm}^2$$

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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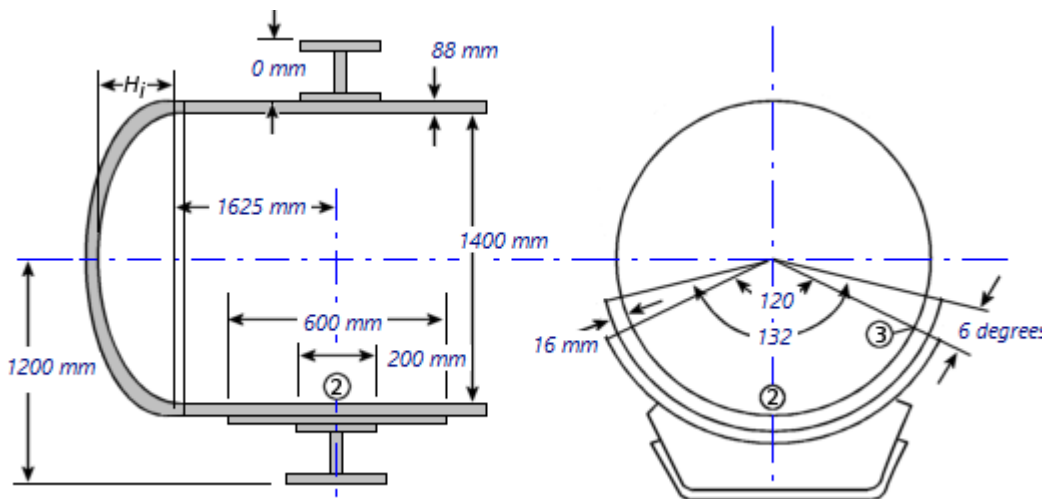


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	745.59	mm
Shell Thickness used in this Case	t	88.000	mm
Stiffened Vessel Length per 4.15.6	L	655.08	cm
Distance from Saddle to Vessel tangent	a1 or a	1625.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1630.99	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1589.87	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		2008.94	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	66357.92	kgf
Pressure used in Analysis	P	0.000 kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-2.47	-2518.57
Long. Stress at Top of Midspan	2.47	1407.13
Long. Stress at Bottom of Midspan	2.47	1407.13
Long. Stress at Top of Saddles	151.54	1407.13
Long. Stress at Bottom of Saddles	-83.99	-2518.57
Long. Stress at Bottom of Saddles	83.99	1407.13
Tangential Shear in Shell	61.89	1125.70
Circ. Stress at Horn of Saddle	79.53	2110.69



Circ. Compressive Stress in Shell	8.45	1407.13
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Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:

$$= \text{Bundle Pull Out Force} \cdot \text{Impact Factor} \cdot B / L_s$$

$$= 2009 \cdot 1.5 \cdot 1200/3250$$

$$= 1112.6 \text{ kgf}$$

Load Combination Results for Tube Bundle Pull Out [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 65245 + \max(1113, 0, 0, 0)$$

$$= 66357.9 \text{ kgf}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)

66747.09 kgf

Transverse Shear Load

0.00 kgf

Longitudinal Shear Load

3013.41 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1260.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0529 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923 \quad K6p = 0.0434$$

$$K7p = 0.0434$$

*The suffix 'p' denotes the values for a wear plate if it exists.***Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -66358 \cdot 162 [1 - (1 - 162/655 + (74.6^2 - 0^2)/(2 \cdot 162 \cdot 655))/(1 + (4 \cdot 0)/(3 \cdot 655))]$$

$$= -23933.3 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 66358 \cdot 655/4 ((1 + 2(74.6^2 - 0^2)/(655^2))/(1 + (4 \cdot 0)/(3 \cdot 655)) - 4 \cdot 162/655)$$

$$= 3658.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 746/(2 \cdot 84.8) - 3658/(\pi \cdot 746^2 \cdot 84.8)$$

$$= -2.47 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 746/(2 \cdot 84.8) + 3658/(\pi \cdot 746^2 \cdot 84.8)$$

$$= 2.47 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 746/(2 \cdot 84.8) - 23933/(0.11 \cdot \pi \cdot 746^2 \cdot 84.8)$$

$$= 151.54 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 746/(2 \cdot 84.8) + 23933/(0.19 \cdot \pi \cdot 746^2 \cdot 84.8)$$

$$= -83.99 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm)/3)$$



$$= 66358 (655 - 2 \cdot 162) / (655 + (4 \cdot 0/3))$$

$$= 33436.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 33436 / (746 \cdot 84.8)$$

$$= 61.89 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(746 \cdot 84.8)}$$

$$= 196.158 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B_1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(746 \cdot 84.8)}, 2 \cdot 162)$$

$$= 592.32 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1631/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 66358 \cdot 0.1 / (592(84.8 + 1 \cdot 16))$$

$$= -8.45 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -66358 / (4(84.8 + 1 \cdot 16)592) -$$

$$3 \cdot 0.053 \cdot 66358 / (2(84.8 + 1 \cdot 16)^2)$$

$$= -79.53 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(703 \cdot 84.8)} + 600)^3) / 12$$

$$= 4989.492 \text{ cm}^4$$

Resolved Inertia for the Shell [I_{shell}]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 4989 + 832(96.9 - 42.4)^2$$

$$= 29712.166 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	981.0	84.8	42.4	832.1	3529263.5	0.297E+05
Wearplate	600.0	16.0	92.8	96.0	891119.9	36.6
Web	10.0	362.0	281.8	36.2	1020206.6	0.163E+05
BasePlate	300.0	34.0	479.8	102.0	4894215.0	0.150E+06
Totals	...	...	...	1066.3	10334805.0	0.196E+06

Distance to Centroid [C_1]:

$$= AY / A$$

$$= 10334805/1066$$



$$= 96.920 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 66358$$

$$= 13505.2764 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 57.6657 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1200 - 703 \cdot \sin(120/2) / 1.05$$

$$= 618.479 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8352.7393 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 41.3611 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(66358 + 389) \cdot 300) / (4 \cdot 1260 \cdot 1767)}$$

$$= 25.969 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (700 + 88 + 16)$$

$$= 1392.569 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1393 / (5 - 1)$$

$$= 348.142 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 348 \cdot 300 / 2$$

$$= 522.213 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 66747 / (1260 \cdot 300)$$

$$= 17.658 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 522 \cdot 17.7$$

$$= 9221.222 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 17.4$$

$$= 36.407 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 9221/36.4$$



= 253.281 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 200 - 10
 = 190.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/36.4
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 200/2
 = 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{667/36.4}$
 = 42.792 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 1200 - cos(120/2)(700 + 88 + 16) - 34
 = 764.000 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120/2)(700 + 88 + 16)
 = 1392.569 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 1260 - 50.8
 = 1209.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (1393 - 1209)/2
 = 91.684 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{764^2 + 91.7^2}$
 = 769.482 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2024850/2651}$
 = 122.785

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 769/42.8
 = 17.982

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 3013 / (2 • 1260) • 348 • 769/2
 = 160.171 kgf-m.



Compressive Allowable, $KL/r < C_c$ ($18 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (18)^2 / (2 \cdot 123^2)) 2651 / \\ &\quad (5/3 + 3 \cdot (18) / (8 \cdot 123) - (18^3) / (8 \cdot 123^3)) \\ &= 1524 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\ &= 253 / 1524 + (160 \cdot 100 / 6666666) / 1767 \\ &= 0.302 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	348.1	10.0	0.0	34.8	0.0	2.90
Totals	...	...	...	53.8	...	669.

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0 / 53.8 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 200 / 2 \\ &= 100.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \text{Outside Radius} - B_{pthk} \\ &= 1200 - \cos(804/2) (34 + 0 + 0) - 0 \\ &= 362.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\ &= \sqrt{ (669 / 53.8) } \\ &= 35.271 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 362 / 35.3 \\ &= 10.263 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} &= F_l / (2 \cdot \text{Baseplate Length}) \\ &= 3013 / (2 \cdot 1260) \\ &= 1.196 \text{ kgf/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} &= \text{Unit Force} \cdot e \cdot L \\ &= 1.2 \cdot 348 \cdot 362 \\ &= 150.704 \text{ kgf-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} &= \text{Bending Moment} / \text{Section Modulus} \\ &= 151 / 66948 \\ &= 225.104 \text{ kgf/cm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($10.3 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (10.3)^2 / (2 \cdot 123^2)) 2651 / \\ &\quad (5/3 + 3 \cdot (10.3) / (8 \cdot 123) - (10.3^3) / (8 \cdot 123^3)) \\ &= 1556 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba} \\ &= 301 / 1556 + (151 \cdot 100 / 669) / 1767 \\ &= 0.321 \end{aligned}$$



Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	5.4000 cm ²
Saddle Load Q0 (Weight)	Q0	66747.1 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3013.4 kgf
Maximum Transverse Force	Ft	0.0 kgf
Maximum Longitudinal Force	Fl	3013.4 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3013 / (5.4 \cdot 4)$$

$$= 139.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (5.4 \cdot 4)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3013 / (1052 \cdot 4)$$

$$= 0.7159 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1200 \cdot 0 + 0$$

$$= 0.00 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0 / 66747$$

$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.72, 0]$$

$$= 0.7159 \text{ cm}^2$$

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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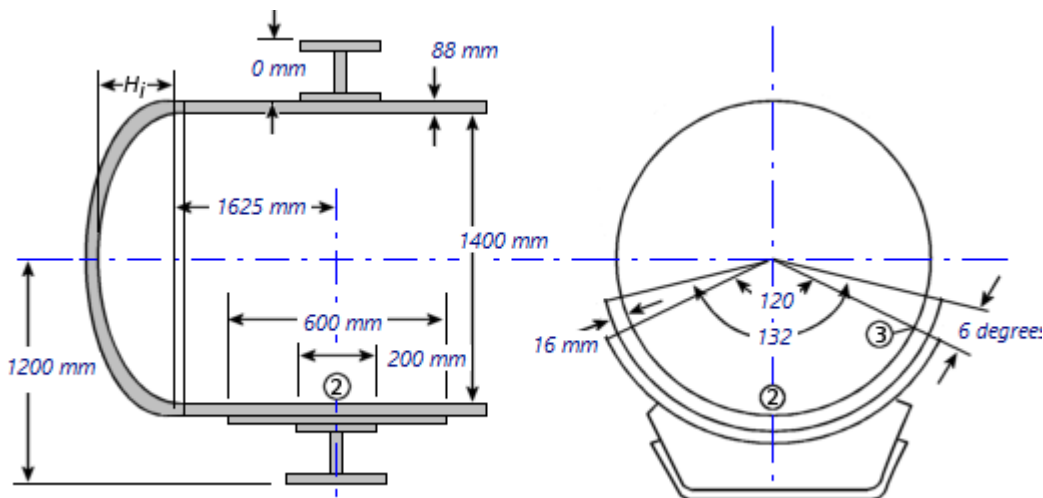


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	745.59	mm
Shell Thickness used in this Case	t	88.000	mm
Stiffened Vessel Length per 4.15.6	L	655.08	cm
Distance from Saddle to Vessel tangent	a1 or a	1625.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1630.99	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1589.87	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		94100.53	kgf
Pressure used in Analysis	P	157.558	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		688.94	1407.13
Long. Stress at Bottom of Midspan		695.95	1407.13
Long. Stress at Top of Saddles		907.34	1407.13
Long. Stress at Bottom of Saddles		573.34	1407.13
Tangential Shear in Shell		87.77	1125.70
Circ. Stress at Horn of Saddle		112.78	1758.91
Circ. Compressive Stress in Shell		11.98	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$\begin{aligned} &= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\ &= 3 (3018/2 + 0) \cdot 1200/1260 \\ &= 4311.6 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Saddle Reaction Force [Fwl]:} \\ &= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / Ls \\ &= \max(589, 0) \cdot 1200/3250 \\ &= 217.3 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Saddle Reaction Force due to Earthquake Fl [Fsl]:} \\ &= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / Ls \\ &= \max(10395, 0) \cdot 1200/3250 \\ &= 3838.1 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Saddle Reaction Force due to Earthquake Ft [Fst]:} \\ &= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\ &= 3 (10395/2 + 0) \cdot 1200/1260 \\ &= 14849.9 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Load Combination Results for Q + Wind or Seismic [Q]:} \\ &= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst}) \\ &= 79251 + \max(217, 4312, 3838, 14850) \\ &= 94100.5 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Wind Force [Fl]:} \\ &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\ &= 1.1 \cdot 1868 (0 + \pi/4(1.58 \cdot 1.2)^2) \\ &= 5771.690 \text{ N} \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	94489.70	kgf
Transverse Shear Load	5197.48	kgf
Longitudinal Shear Load	10394.96	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

$$\begin{aligned} &\text{Saddle Dimension [E]:} \\ &= \text{Baseplate Length} \\ &= 1260.000 \text{ mm} \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

$$\begin{aligned} &\text{Moment per Equation 4.15.1 [M1]:} \\ &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)] \\ &= -94101 \cdot 162 [1 - (1 - 162/655 + (74.6^2 - 0^2) / (2 \cdot 162 \cdot 655)) / (1 + (4 \cdot 0)/(3 \cdot 655))] \\ &= -33939.2 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Moment per Equation 4.15.2 [M2]:} \\ &= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L) \\ &= 94101 \cdot 655/4 ((1 + 2(74.6^2 - 0^2)/(655^2)) / (1 + (4 \cdot 0)/(3 \cdot 655)) - 4 \cdot 162/655) \\ &= 5187.8 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Top of Shell (4.15.4) [\sigma_1]:} \\ &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\ &= 158 \cdot 746/(2 \cdot 84.8) - 5188/(\pi \cdot 746^2 \cdot 84.8) \\ &= 688.94 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Bottom of Shell (4.15.5) [\sigma_2]:} \\ &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\ &= 158 \cdot 746/(2 \cdot 84.8) + 5188/(\pi \cdot 746^2 \cdot 84.8) \end{aligned}$$



= 695.95 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 158 \cdot 746 / (2 \cdot 84.8) - 33939 / (0.11 \cdot \pi \cdot 746^2 \cdot 84.8)$$

$$= 907.34 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 158 \cdot 746 / (2 \cdot 84.8) + 33939 / (0.19 \cdot \pi \cdot 746^2 \cdot 84.8)$$

$$= 573.34 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 94101 (655 - 2 \cdot 162) / (655 + (4 \cdot 0 / 3))$$

$$= 47415.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 47415 / (746 \cdot 84.8)$$

$$= 87.77 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(746 \cdot 84.8)}$$

$$= 196.158 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(746 \cdot 84.8)}, 2 \cdot 162)$$

$$= 592.32 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1631 / 1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 94101 \cdot 0.1 / (592(84.8 + 1 \cdot 16))$$

$$= -11.98 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -94101 / (4(84.8 + 1 \cdot 16)592) -$$

$$3 \cdot 0.053 \cdot 94101 / (2(84.8 + 1 \cdot 16)^2)$$

$$= -112.78 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3250.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12059\text{E-}04 \cdot 3250 (93.3 - 21.1)$$

$$= 2.830 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2450.6	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(703 \cdot 84.8)} + 600)^3) / 12$$

$$= 4989.492 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 4989 + 832 (96.9 - 42.4)^2$$

$$= 29712.166 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	981.0	84.8	42.4	832.1	3529263.5	0.297E+05
Wearplate	600.0	16.0	92.8	96.0	891119.9	36.6
Web	10.0	362.0	281.8	36.2	1020206.6	0.163E+05
BasePlate	300.0	34.0	479.8	102.0	4894215.0	0.150E+06
Totals	...	...	...	1066.3	10334805.0	0.196E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10334805 / 1066$$

$$= 96.920 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 94101$$

$$= 19151.4980 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 81.7742 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1470.3682 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1200 - 703 \cdot \sin(120/2) / 1.05$$

$$= 618.479 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 11844.8125 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 58.6532 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1633.7424 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(94101 + 389) \cdot 300) / (4 \cdot 1260 \cdot 1634)}$$

$$= 32.137 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (700 + 88 + 16)$$

$$= 1392.569 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1393 / (5 - 1)$$

$$= 348.142 \text{ mm}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 348 \cdot 300 / 2$$

$$= 522.213 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 94490 / (1260 \cdot 300)$$

$$= 24.997 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 522 \cdot 25$$

$$= 13053.909 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 19 + 17.4$$

$$= 36.407 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 13054 / 36.4$$

$$= 358.554 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 36.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (667 / 36.4) }$$

$$= 42.792 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 1200 - \cos(120 / 2)(700 + 88 + 16) - 34$$

$$= 764.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120 / 2)(700 + 88 + 16)$$

$$= 1392.569 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 1260 - 50.8$$

$$= 1209.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$



$$= (1393 - 1209) / 2$$

$$= 91.684 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{764^2 + 91.7^2}$$

$$= 769.482 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2024850 / 2451)}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 769 / 42.8$$

$$= 17.982$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 10395 / (2 \cdot 1260) \cdot 348 \cdot 769 / 2$$

$$= 552.520 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18)^2 / (2 \cdot 128^2)) 2451 /$$

$$(5/3 + 3 \cdot (18) / (8 \cdot 128) - (18^3) / (8 \cdot 128^3))$$

$$= 1411 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 359 / 1411 + (553 \cdot 100 / 6666666) / 1634$$

$$= 0.761$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	348.1	10.0	0.0	34.8	0.0	2.90
Totals	...	...	...	53.8	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 53.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1200 - \cos(804/2) (34 + 0 + 0) - 0$$

$$= 362.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{669 / 53.8}$$

$$= 35.271 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 362 / 35.3$$

$$= 10.263$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 10395 / (2 \cdot 1260)$$

$$= 4.125 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.12 \cdot 348 \cdot 362$$

$$= 519.862 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 520 / 66948$$

$$= 776.511 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.3 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.3)^2 / (2 \cdot 128^2)) 2451 /$$

$$(5/3 + 3 \cdot (10.3) / (8 \cdot 128) - (10.3^3) / (8 \cdot 128^3))$$

$$= 1440 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 426 / 1440 + (520 \cdot 100 / 669) / 1634$$

$$= 0.771$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.02 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1630.99 kgf/cm ²
Area Available in a Single Bolt	BltArea	5.4000 cm ²
Saddle Load QO (Weight)	QO	79639.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3838.1 kgf
Maximum Transverse Force	Ft	5197.5 kgf
Maximum Longitudinal Force	Fl	10395.0 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 2153.6 \text{ kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 10395 / (5.4 \cdot 4)$$

$$= 481.2 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5197 / (5.4 \cdot 4)$$

$$= 240.6 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 10395 / (1100 \cdot 4)$$

$$= 2.3626 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1200 \cdot 5197 + 0$$



= 6236.98 kgf-m.

Eccentricity (e):

= Rmom / QO

= 6237/79640

= 78.31 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 2.36, 0]

= 2.3626 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -13 °C

User entered Minimum Design Temperature : -29 °C

The Curve for this Material is : B

From Figure 3.8M- with PWHT, the MDMT is : -13 °C

Temperature reduction per Graph 3.13 : 0 °C

The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

= (tr • E*)/(tnom - ci - cext)

= (83.3 • 1)/(16 - 3.17 - 0)

= 1.000

Warning: Impact Testing Is Mandatory!

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	745.59	mm
Shell Thickness used in this Case	t	88.000	mm
Stiffened Vessel Length per 4.15.6	L	655.08	cm
Distance from Saddle to Vessel tangent	a1 or a	1625.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1630.99	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	38.96	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		21639.51	kgf
Pressure used in Analysis	P	157.558	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		693.60	1407.13
Long. Stress at Bottom of Midspan		691.29	1407.13



Long. Stress at Top of Saddles	756.06	1407.13
Long. Stress at Bottom of Saddles	657.19	1407.13
Tangential Shear in Shell	18.70	1125.70
Circ. Stress at Horn of Saddle	25.93	1758.91
Circ. Compressive Stress in Shell	2.75	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (3018/2 + 0) \cdot 1200/1260$$

$$= 4311.6 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(589, 0) \cdot 1200/3250$$

$$= 217.3 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(10395, 0) \cdot 1200/3250$$

$$= 3838.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (10395/2 + 0) \cdot 1200/1260$$

$$= 14849.9 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 6790 + \max(217, 4312, 3838, 14850)$$

$$= 21639.5 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1868 (0 + \pi/4 (1.58 \cdot 1.2)^2)$$

$$= 5771.690 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22028.67	kgf
Transverse Shear Load	5197.48	kgf
Longitudinal Shear Load	10394.96	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1260.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -21640 \cdot 162 [1 - (1 - 162/655 + (74.6^2 - 39^2)/$$

$$(2 \cdot 162 \cdot 655)) / (1 + (4 \cdot 39)/(3 \cdot 655))]$$

$$= -10047.1 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 21640 \cdot 655/4 ((1 + 2(74.6^2 - 39^2)/(655^2)) / (1 + (4 \cdot 39)/$$



$$(3 \cdot 655) - 4 \cdot 162 / 655 \\ = -1710.4 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 158 \cdot 746 / (2 \cdot 84.8) - 1710 / (\pi \cdot 746^2 \cdot 84.8) \\ = 693.60 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 158 \cdot 746 / (2 \cdot 84.8) + 1710 / (\pi \cdot 746^2 \cdot 84.8) \\ = 691.29 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 158 \cdot 746 / (2 \cdot 84.8) - 10047 / (0.11 \cdot \pi \cdot 746^2 \cdot 84.8) \\ = 756.06 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 158 \cdot 746 / (2 \cdot 84.8) + 10047 / (0.19 \cdot \pi \cdot 746^2 \cdot 84.8) \\ = 657.19 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 21640 (655 - 2 \cdot 162) / (655 + (4 \cdot 39 / 3)) \\ = 10102.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 10103 / (746 \cdot 84.8) \\ = 18.70 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{746 \cdot 84.8} \\ = 196.158 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(200 + 1.56 \cdot \sqrt{746 \cdot 84.8}, 2 \cdot 162) \\ = 592.32 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(1631 / 1407, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r)) \\ = -0.76 \cdot 21640 \cdot 0.1 / (592(84.8 + 1 \cdot 16)) \\ = -2.75 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2) \\ = -21640 / (4(84.8 + 1 \cdot 16) 592) - \\ 3 \cdot 0.053 \cdot 21640 / (2(84.8 + 1 \cdot 16)^2) \\ = -25.93 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2450.6	kgf/cm ²



Height of Web at Center Hw,c 500.0 mm
 Friction Coefficient mu 0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(703 \cdot 84.8)} + 600)^3) / 12$
 $= 4989.492 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 4989 + 832 (96.9 - 42.4)^2$
 $= 29712.166 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	981.0	84.8	42.4	832.1	3529263.5	0.297E+05
Wearplate	600.0	16.0	92.8	96.0	891119.9	36.6
Web	10.0	362.0	281.8	36.2	1020206.6	0.163E+05
BasePlate	300.0	34.0	479.8	102.0	4894215.0	0.150E+06
Totals	...	...	...	1066.3	10334805.0	0.196E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 10334805 / 1066$
 $= 96.920 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 21640$
 $= 4404.1104 \text{ kgf}$

Tension Stress, St = (Fh/As) = 18.8049 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1470.3682 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 1200 - 703 \cdot \sin(120 / 2) / 1.05$
 $= 618.479 \text{ mm}$

Bending Moment, M = Fh * d = 2723.8528 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 13.4880 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1633.7424 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(21640 + 389)300 / (4 \cdot 1260 \cdot 1634)}$
 $= 15.517 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120 / 2) (700 + 88 + 16)$
 $= 1392.569 \text{ mm}$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 1393 / (5 - 1) \\
 &= 348.142 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 348 \cdot 300 / 2 \\
 &= 522.213 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 22029 / (1260 \cdot 300) \\
 &= 5.828 \text{ kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 522 \cdot 5.83 \\
 &= 3043.298 \text{ kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 19 + 17.4 \\
 &= 36.407 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 3043 / 36.4 \\
 &= 83.591 \text{ kgf/cm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 200 - 10 \\
 &= 190.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0 / 36.4 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 200 / 2 \\
 &= 100.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\
 &= \sqrt{ (667 / 36.4) } \\
 &= 42.792 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1200 - \cos(120 / 2) (700 + 88 + 16) - 34 \\
 &= 764.000 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120 / 2) (700 + 88 + 16) \\
 &= 1392.569 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1260 - 50.8
 \end{aligned}$$



= 1209.200 mm

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (1393 - 1209) / 2$$

$$= 91.684 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{764^2 + 91.7^2}$$

$$= 769.482 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2024846 / 2451)}$$

$$= 127.710$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 769 / 42.8$$

$$= 17.982$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 10395 / (2 \cdot 1260) \cdot 348 \cdot 769 / 2$$

$$= 552.520 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($18 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (18)^2 / (2 \cdot 128^2)) 2451 /$$

$$(5/3 + 3 \cdot (18) / (8 \cdot 128) - (18^3) / (8 \cdot 128^3))$$

$$= 1411 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 83.6 / 1411 + (553 \cdot 100 / 66666666) / 1634$$

$$= 0.567$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	348.1	10.0	0.0	34.8	0.0	2.90
Totals	...	...	...	53.8	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 53.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1200 - \cos(804/2) (34 + 0 + 0) - 0$$

$$= 362.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669 / 53.8}$$

$$= 35.271 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 362 / 35.3$$



= 10.263

Unit Force [Force,u]:
 = $F_l / (2 \cdot \text{Baseplate Length})$
 = $10395 / (2 \cdot 1260)$
 = 4.125 kgf/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force $\cdot e \cdot L$
 = $4.12 \cdot 348 \cdot 362$
 = 519.862 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $520/66948$
 = 776.511 kgf/cm²

Compressive Allowable, $KL/r < C_c$ ($10.3 < 128$) per AISC E2-1 [Sca]:
 = $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (10.3)^2 / (2 \cdot 128^2)) 2451 /$
 $(5/3 + 3 \cdot (10.3) / (8 \cdot 128) - (10.3^3) / (8 \cdot 128^3))$
 = 1440 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 = $99.3/1440 + (520 \cdot 100/669) / 1634$
 = 0.544

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.02 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1630.99 kgf/cm ²
Area Available in a Single Bolt	BltArea	5.4000 cm ²
Saddle Load QO (Weight)	QO	7178.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3838.1 kgf
Maximum Transverse Force	Ft	5197.5 kgf
Maximum Longitudinal Force	Fl	10395.0 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $10395 / (5.4 \cdot 4)$
 = 481.2 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $5197 / (5.4 \cdot 4)$
 = 240.6 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $10395 / (1100 \cdot 4)$
 = 2.3626 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$



$$= 1200 \cdot 5197 + 0$$

$$= 6236.98 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 6237 / 7179$$

$$= 868.81 \text{ mm} > B_{plen} / 6 \rightarrow \text{Uplift in Transverse direction}$$

$$f = B_{plen} / 2 - Edgedis$$

$$= 1260 / 2 - 50.8$$

$$= 579.20 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_S / E_C$$

$$= 2074056 / 219500$$

$$= 9.45$$

$$K1 = 3 (e - 0.5 \cdot B_{plen})$$

$$= 3 (869 - 0.5 \cdot 1260)$$

$$= 716.44 \text{ mm}$$

$$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$$

$$= 6 \cdot 9.45 \cdot 10.8 / 300 \cdot (579 + 869)$$

$$= 295537.59 \text{ mm}^2$$

$$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$$

$$= -295538 \cdot (0.5 \cdot 1260 + 579)$$

$$= -357364043.20 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + 716 \cdot Y^2 + 295538 \cdot Y + -357364032 = 0$$

$$Y = 442.28 \text{ mm}$$

$$Num = (B_{plen} / 2 - Y / 3 - e)$$

$$= (1260 / 2 - 442 / 3 - 869)$$

$$= -386.24$$

$$Denom = (B_{plen} / 2 - Y / 3 + f)$$

$$= (1260 / 2 - 442 / 3 + 579)$$

$$= 1061.77$$

Total Bolt Tension Force [Tforce]:

$$= -QO \cdot Num / Denom$$

$$= -7179 \cdot -386 / 1062$$

$$= 2611.39 \text{ kgf}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= Tforce / (Stba \cdot Nbt)$$

$$= 2611 / (1400 \cdot 2)$$

$$= 0.9326 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 2.36, 0.93]$$

$$= 2.3626 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + Tforce) / (Y \cdot B_{pwid})$$

$$= 2(7179 + 2611) / (442 \cdot 300)$$

$$= 14.76 \text{ kgf/cm}^2$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - WebIngh) / 2$$

$$= (1260 - 1209) / 2$$

$$= 25.4000 \text{ mm}$$



Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= T_{\text{force}} \cdot A_{\text{dist}} \\ &= 2611 \cdot 2.54 \\ &= 66.33 \text{ kgf-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= f_c \cdot (Y - A_{\text{dist}}) / Y \\ &= 14.8 \cdot (442 - 25.4) / 442 \\ &= 13.91 \text{ kgf/cm}^2 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (A_{\text{dist}}^2 \cdot B_{\text{pwid}} / 6) \cdot (f_1 + 2 \cdot f_c) \\ &= (25.4^2 \cdot 300 / 6) \cdot (13.9 + 2 \cdot 14.8) \\ &= 14.01 \text{ kgf-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(M_t, M_c) / (B_{\text{pwid}} \cdot S_{ba}))} \\ &= \sqrt{(6 \cdot \max(66.3, 14) / (300 \cdot 2446))} \\ &= 7.3637 \text{ mm} \end{aligned}$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-13 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-13 °C
Temperature reduction per Graph 3.13	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}}) \\ &= (83.3 \cdot 1) / (16 - 3.17 - 0) \\ &= 1.000 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Review notes about nozzle loadings and supports in the Warnings report.

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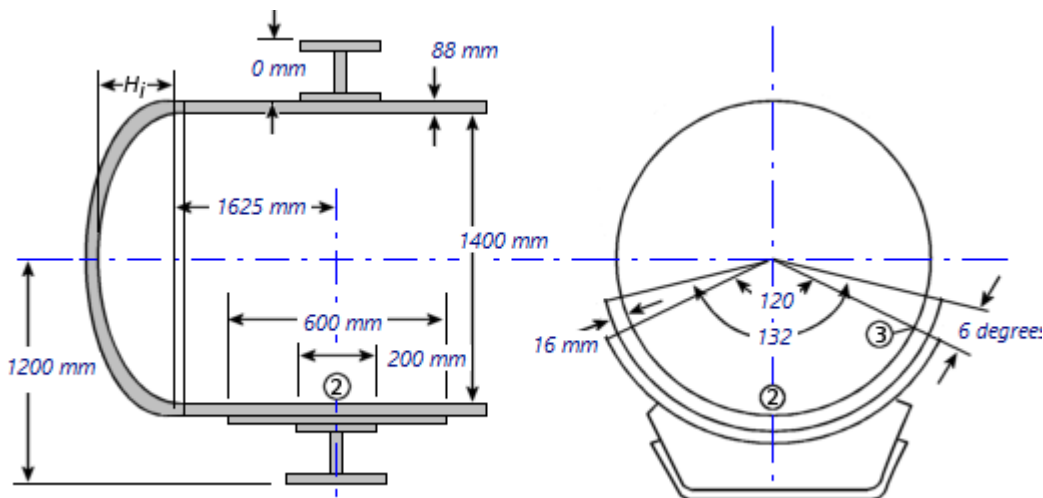


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	744.00	mm
Shell Thickness used in this Case	t	88.000	mm
Stiffened Vessel Length per 4.15.6	L	655.08	cm
Distance from Saddle to Vessel tangent	a1 or a	1625.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1630.99	kgf/cm ²
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		78048.07	kgf
Pressure used in Analysis	P	197.031	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		830.10	2518.57
Long. Stress at Bottom of Midspan		835.71	2518.57
Long. Stress at Top of Saddles		1005.53	2518.57
Long. Stress at Bottom of Saddles		737.22	2518.57
Tangential Shear in Shell		70.32	1413.05
Circ. Stress at Horn of Saddle		97.06	3777.85
Circ. Compressive Stress in Shell		10.07	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (996/2 + 0) \cdot 1200/1260$$

$$= 1422.8 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{F}_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(194, 0) \cdot 1200/3250$$

$$= 71.7 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 76625 + \max(71.7, 1423, 0, 0)$$

$$= 78048.1 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1868 (0 + \pi/4(1.58 \cdot 1.2)^2)$$

$$= 5771.690 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	78437.23	kgf
Transverse Shear Load	497.99	kgf
Longitudinal Shear Load	194.22	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1260.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -78048 \cdot 162 [1 - (1 - 162/655 + (74.4^2 - 0^2)/(2 \cdot 162 \cdot 655))/(1 + (4 \cdot 0)/(3 \cdot 655))]$$

$$= -28163.7 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 78048 \cdot 655/4((1 + 2(74.4^2 - 0^2)/(655^2))/(1 + (4 \cdot 0)/(3 \cdot 655)) - 4 \cdot 162/655)$$

$$= 4288.7 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 197 \cdot 744/(2 \cdot 88) - 4289/(\pi \cdot 744^2 \cdot 88)$$

$$= 830.10 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 197 \cdot 744/(2 \cdot 88) + 4289/(\pi \cdot 744^2 \cdot 88)$$

$$= 835.71 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 197 \cdot 744/(2 \cdot 88) - 28164/(0.11 \cdot \pi \cdot 744^2 \cdot 88)$$

$$= 1005.53 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 197 \cdot 744/(2 \cdot 88) + 28164/(0.19 \cdot \pi \cdot 744^2 \cdot 88)$$



$$= 737.22 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 78048 (655 - 2 \cdot 162)/(655 + (4 \cdot 0/3))$$

$$= 39326.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 39327 / (744 \cdot 88)$$

$$= 70.32 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(744 \cdot 88)}$$

$$= 199.582 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(744 \cdot 88)}, 2 \cdot 162)$$

$$= 599.16 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1631/2519, 1)$$

$$= 0.6476$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 78048 \cdot 0.1 / (599(88 + 0.65 \cdot 16))$$

$$= -10.07 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -78048 / (4(88 + 0.65 \cdot 16)599) -$$

$$3 \cdot 0.053 \cdot 78048 / (2(88 + 0.65 \cdot 16)^2)$$

$$= -97.06 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(700 \cdot 88)} + 600)88^3)/12$$

$$= 5606.134 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 5606 + 869(97 - 44)^2$$

$$= 30041.402 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	987.2	88.0	44.0	868.7	3822367.5	0.300E+05
Wearplate	600.0	16.0	96.0	96.0	921599.9	21.5
Web	10.0	362.0	285.0	36.2	1031700.1	0.167E+05
BasePlate	300.0	34.0	483.0	102.0	4926600.0	0.152E+06
Totals	...	...	...	1102.9	10702268.0	0.199E+06



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 10702268/1103 \\ &= 97.036 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 78048 \\ &= 15884.4746 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 67.8245 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1200 - 700 \cdot \sin(120/2) / 1.05 \\ &= 621.105 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 9865.9316 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 48.1438 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(78048 + 389)300 / (4 \cdot 1260 \cdot 1767)} \\ &= 28.151 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (700 + 88 + 16) \\ &= 1392.569 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1393 / (5 - 1) \\ &= 348.142 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 348 \cdot 300 / 2 \\ &= 522.213 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 78437 / (1260 \cdot 300) \\ &= 20.751 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 522 \cdot 20.8 \\ &= 10836.235 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 19 + 17.4 \\ &= 36.407 \text{ cm}^2 \end{aligned}$$



Compressive Stress [Sc]:

$$\begin{aligned} &= P/A_r \\ &= 10836/36.4 \\ &= 297.641 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 200 - 10 \\ &= 190.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0/36.4 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 200/2 \\ &= 100.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(667/36.4)} \\ &= 42.792 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned} &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\ &= 1200 - \cos(120/2)(700 + 88 + 16) - 34 \\ &= 764.000 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned} &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\ &= 2 \cdot \sin(120/2)(700 + 88 + 16) \\ &= 1392.569 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned} &= \text{Baseplate length} - \text{clearance} \\ &= 1260 - 50.8 \\ &= 1209.200 \text{ mm} \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= (Lw - Bd) / 2 \\ &= (1393 - 1209) / 2 \\ &= 91.684 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{Lv^2 + Dd^2} \\ &= \sqrt{764^2 + 91.7^2} \\ &= 769.482 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 2024850/2651)} \\ &= 122.785 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 769/42.8 \\ &= 17.982 \end{aligned}$$

Bending Moment [Rm]:



$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 194 / (2 \cdot 1260) \cdot 348 \cdot 769 / 2$$

$$= 10.323 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($18 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (18)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (18) / (8 \cdot 123) - (18^3) / (8 \cdot 123^3))$$

$$= 1524 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 298 / 1524 + (10.3 \cdot 100 / 6666666) / 1767$$

$$= 0.204$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	348.1	10.0	0.0	34.8	0.0	2.90
Totals	...	...	...	53.8	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 53.8$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1200 - \cos(804 / 2) (34 + 0 + 0) - 0$$

$$= 362.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (669 / 53.8) }$$

$$= 35.271 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 362 / 35.3$$

$$= 10.263$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 194 / (2 \cdot 1260)$$

$$= 0.077 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 348 \cdot 362$$

$$= 9.713 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.71 / 66948$$

$$= 14.508 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($10.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (10.3)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (10.3) / (8 \cdot 123) - (10.3^3) / (8 \cdot 123^3))$$

$$= 1556 \text{ kgf/cm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 354/1556 + (9.71 \cdot 100/669)/1767$$

$$= 0.236$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BlArea	5.4000 cm ²
Saddle Load QO (Weight)	QO	77014.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	71.7 kgf
Maximum Transverse Force	Ft	498.0 kgf
Maximum Longitudinal Force	Fl	194.2 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 194 / (5.4 \cdot 4)$$

$$= 9.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 498 / (5.4 \cdot 4)$$

$$= 23.1 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $194 / (1052 \cdot 4)$
 = 0.0461 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1200 \cdot 498 + 0$
 = 597.59 kgf-m.

Eccentricity (e):
 = $Rmom / QO$
 = $598 / 77014$
 = 7.76 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 = $\max[0, 0.046, 0]$
 = 0.0461 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	744.00	mm
Shell Thickness used in this Case	t	88.000	mm
Stiffened Vessel Length per 4.15.6	L	655.08	cm



Distance from Saddle to Vessel tangent	a1 or a	1625.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	600.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1630.99	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	38.80	cm
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		5069.79	kgf
Pressure used in Analysis	P	197.031	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	833.16	2518.57
Long. Stress at Bottom of Midspan	832.64	2518.57
Long. Stress at Top of Saddles	847.32	2518.57
Long. Stress at Bottom of Saddles	824.91	2518.57
Tangential Shear in Shell	4.23	1413.05
Circ. Stress at Horn of Saddle	6.30	3777.85
Circ. Compressive Stress in Shell	0.65	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (996/2 + 0) \cdot 1200/1260$
 $= 1422.8\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(194, 0) \cdot 1200/3250$
 $= 71.7\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 3647 + \max(71.7, 1423, 0, 0)$
 $= 5069.8\ kgf$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1868 (0 + \pi/4(1.58 \cdot 1.2)^2)$
 $= 5771.690\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5458.95	kgf
Transverse Shear Load	497.99	kgf
Longitudinal Shear Load	194.22	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 1260.000\ mm$



The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \frac{(1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))}{(1 + (4h_m)/3L)} \right]$$

$$= -5070 \cdot 162 \left[1 - \frac{(1 - 162/655 + (74.4^2 - 38.8^2)/(2 \cdot 162 \cdot 655))}{(1 + (4 \cdot 38.8)/(3 \cdot 655))} \right]$$

$$= -2352.5 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \left(\frac{(1 + 2(R_m^2 - h_m^2)/(L^2))}{(1 + (4h_m^2)/(3L))} - 4a/L \right)$$

$$= 5070 \cdot 655/4 \left(\frac{(1 + 2(74.4^2 - 38.8^2)/(655^2))}{(1 + (4 \cdot 38.8)/(3 \cdot 655))} - 4 \cdot 162/655 \right)$$

$$= -398.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 197 \cdot 744/(2 \cdot 88) - 399/(\pi \cdot 744^2 \cdot 88)$$

$$= 833.16 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 197 \cdot 744/(2 \cdot 88) + 399/(\pi \cdot 744^2 \cdot 88)$$

$$= 832.64 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 197 \cdot 744/(2 \cdot 88) - 2353/(0.11 \cdot \pi \cdot 744^2 \cdot 88)$$

$$= 847.32 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 197 \cdot 744/(2 \cdot 88) + 2353/(0.19 \cdot \pi \cdot 744^2 \cdot 88)$$

$$= 824.91 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 5070 (655 - 2 \cdot 162)/(655 + (4 \cdot 38.8/3))$$

$$= 2367.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 2368/(744 \cdot 88)$$

$$= 4.23 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(744 \cdot 88)}$$

$$= 199.582 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(744 \cdot 88)}, 2 \cdot 162)$$

$$= 599.16 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1631/2519, 1)$$

$$= 0.6476$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$



$$= -0.76 \cdot 5070 \cdot 0.1 / (599 (88 + 0.65 \cdot 16))$$

$$= -0.65 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -5070 / (4(88 + 0.65 \cdot 16)599) -$$

$$3 \cdot 0.053 \cdot 5070 / (2(88 + 0.65 \cdot 16)^2)$$

$$= -6.30 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1260.0000	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	500.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(700 \cdot 88) + 600})^3) / 12$$

$$= 5606.134 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 5606 + 869 (97 - 44)^2$$

$$= 30041.402 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	987.2	88.0	44.0	868.7	3822367.5	0.300E+05
Wearplate	600.0	16.0	96.0	96.0	921599.9	21.5
Web	10.0	362.0	285.0	36.2	1031700.1	0.167E+05
BasePlate	300.0	34.0	483.0	102.0	4926600.0	0.152E+06
Totals	...	...	...	1102.9	10702268.0	0.199E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10702268 / 1103$$

$$= 97.036 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 5070$$

$$= 1031.8113 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.4057 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1200 - 700 \cdot \sin(120/2) / 1.05$$

$$= 621.105 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 640.8635 \text{ kgf-m.}$$

$$\begin{aligned}\text{Bending Stress, } S_b &= (M \cdot C_1 / I) = 3.1273 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } S_a &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2\end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}&= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\&= \sqrt{(3(5070 + 389) \cdot 300) / (4 \cdot 1260 \cdot 1767)} \\&= 7.427 \text{ mm}\end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}\text{Web Length Dimension [Web Length]:} \\&= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\&= 2 \cdot \cos(90 - 120/2) (700 + 88 + 16) \\&= 1392.569 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Distance between Ribs [e]:} \\&= \text{Web Length} / (\text{Nribs} - 1) \\&= 1393 / (5 - 1) \\&= 348.142 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Baseplate Pressure Area [Ap]:} \\&= e \cdot \text{Bpwid} / 2 \\&= 348 \cdot 300 / 2 \\&= 522.213 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Bearing Pressure [Bp]:} \\&= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\&= 5459 / (1260 \cdot 300) \\&= 1.444 \text{ kgf/cm}^2\end{aligned}$$

$$\begin{aligned}\text{Axial Load [P]:} \\&= A_p \cdot B_p \\&= 522 \cdot 1.44 \\&= 754.162 \text{ kgf}\end{aligned}$$

$$\begin{aligned}\text{Area of the Rib and Web [Ar]:} \\&= \text{Rib Area} + \text{Web Area} \\&= 19 + 17.4 \\&= 36.407 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Compressive Stress [Sc]:} \\&= P / A_r \\&= 754 / 36.4 \\&= 20.715 \text{ kgf/cm}^2\end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

$$\begin{aligned}\text{Rib dimension [D]:} \\&= \text{Saddle Width} - \text{Web Thickness} \\&= 200 - 10 \\&= 190.000 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Distance to Centroid from Datum [ytot]:} \\&= AY / A \\&= 0 / 36.4 \\&= 0.000 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Distance to Centroid [C1]:} \\&= \text{Saddle Width} / 2 \\&= 200 / 2 \\&= 100.000 \text{ mm}\end{aligned}$$



Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{667/36.4}$$

$$= 42.792 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1200 - \cos(120/2)(700 + 88 + 16) - 34$$

$$= 764.000 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2)(700 + 88 + 16)$$

$$= 1392.569 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1260 - 50.8$$

$$= 1209.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (1393 - 1209) / 2$$

$$= 91.684 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{\text{Lv}^2 + \text{Dd}^2}$$

$$= \sqrt{764^2 + 91.7^2}$$

$$= 769.482 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2024846 / 2651)}$$

$$= 122.785$$

Slenderness ratio [KL/r]:

$$= \text{KL} / \text{r}$$

$$= 1 \cdot 769 / 42.8$$

$$= 17.982$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot \text{e} \cdot \text{L} / 2$$

$$= 194 / (2 \cdot 1260) \cdot 348 \cdot 769 / 2$$

$$= 10.323 \text{ kgf-m.}$$

Compressive Allowable, $\text{KL}/\text{r} < \text{Cc}$ ($18 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/\text{r})^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/\text{r}) / (8 \cdot \text{Cc}) - (\text{KL}/\text{r})^2 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (18)^2 / (2 \cdot 123^2)) 2651 / (5/3 + 3 \cdot (18) / (8 \cdot 123) - (18^3) / (8 \cdot 123^3))$$

$$= 1524 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc} / \text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 20.7 / 1524 + (10.3 \cdot 100 / 6666666) / 1767$$

$$= 0.022$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	348.1	10.0	0.0	34.8	0.0	2.90
Totals	...	...	...	53.8	...	669.

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 53.8$$

$$= 0.000 \text{ mm}$$



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1200 - \cos(804/2)(34 + 0 + 0) - 0$$

$$= 362.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(669/53.8)}$$

$$= 35.271 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 362/35.3$$

$$= 10.263$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 194/(2 \cdot 1260)$$

$$= 0.077 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 348 \cdot 362$$

$$= 9.713 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.71/66948$$

$$= 14.508 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($10.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.3)^2 / (2 \cdot 123^2)) 2651 /$$

$$(5/3 + 3 \cdot (10.3) / (8 \cdot 123) - (10.3)^3 / (8 \cdot 123^3))$$

$$= 1556 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 24.6/1556 + (9.71 \cdot 100/669) / 1767$$

$$= 0.024$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	5.4000 cm ²
Saddle Load Q0 (Weight)	Q0	4036.1 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	71.7 kgf
Maximum Transverse Force	Ft	498.0 kgf
Maximum Longitudinal Force	F1	194.2 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 194/(5.4 \cdot 4)$$

$$= 9.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:



= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $498 / (5.4 \cdot 4)$
= 23.1 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_t / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= $194 / (1052 \cdot 4)$
= 0.0461 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1200 \cdot 498 + 0$
= 597.59 kgf-m .

Eccentricity (e):
= R_{mom} / QO
= $598 / 4036$
= $148.06 \text{ mm} < B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.046, 0]$
= 0.0461 cm^2

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N1-Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	121.145	kgf/cm ²
Temperature for Internal Pressure	Temp	66	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1400.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		146.32	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

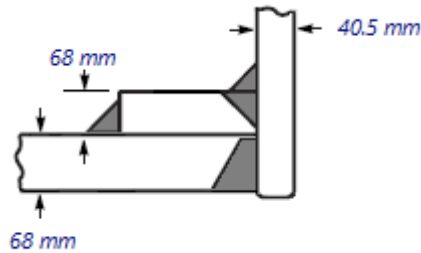
Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	606.4000	mm
Thickness of Pad	te	68.0000	mm
Weld leg size between Pad and Shell	Wp	68.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	68.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1-Tube inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
= $0.5 \cdot 332 \cdot (\exp(121/(1203 \cdot 1)) - 1) + 3.17 + 0$
= $17.6 + 3.17 + 0 = 20.7$ mm Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
= $S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
= $1203 \cdot 1 \cdot \ln((2 \cdot 37.3 + 332) / 332) - 0$
= $244 - 0 = 244$ kgf/cm²

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm
Min. required shell thickness, internal p + c : 66.437 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 35.408 mm , is greater than required thickness 11.515 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 458.2,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2549.8 kgf/cm ²	Passed
Occasional	: 284.2,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 115.2,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
= $\min[S_n/S, 1]$
= $\min[1203/1407, 1]$
= 0.855

Pad Material Factor [frp]:
= $\min[S_p/S, 1]$
= $\min[1407/1407, 1]$
= 1.000

Thickness of Nozzle [tn]:
= thickness - corrosion allowance
= $40.5 - 3.17$
= 37.287 mm



Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 1406/64.8 \\ &= 21.695 \text{ must be less than } 400. \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1400/2 + 3.17 \\ &= 703.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$\begin{aligned} &= \sqrt{(Reff \cdot t) + W} \\ &= \sqrt{(703 \cdot 64.8) + 100} \\ &= 313.503 \text{ mm} \end{aligned}$$

Limit [LR2]:

$$\begin{aligned} &= \sqrt{((Reff + t) \cdot (t + t_e))} \\ &= \sqrt{((703 + 64.8) \cdot (64.8 + 68))} \\ &= 319.389 \text{ mm} \end{aligned}$$

Limit [LR3]:

$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 166 \\ &= 331.826 \text{ mm} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(314, 319, 332) \\ &= 313.503 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 64.8, 68] + \sqrt{(166 \cdot 37.3)} \\ &= 146.654 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(64.8 + 68) \\ &= 1062.600 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\ &= \min[147, 300, 1063] + 64.8 \\ &= 211.479 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 64.8 + ((68 \cdot 1) / 314) \\ &= 86.515 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / \sqrt{((D_i + t_{eff}) \cdot t_{eff})}) \\ &= \min(12, (2 \cdot 166 + 37.3) / \sqrt{((1406 + 86.5) \cdot 86.5)}) \\ &= 1.027 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 64.8 \cdot 314 \cdot \max((1.03/5)^{0.85}, 1) \\ &= 203.228 \text{ cm}^2 \end{aligned}$$



Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 37.3 \cdot 211 \\ &= 78.855 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 68^2 \\ &= 23.120 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(100 \cdot 68, 314 \cdot 68) \\ &= 68.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 203 + 0.86(78.9 + 0) + 0.5 + 23.1 + 0 + 1(68) \\ &= 362.275 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 37.3 / \ln[1 + 37.3/166] \\ &= 183.927 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 86.5 / \ln[1 + 86.5/703] \\ &= 745.596 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1407 \cdot 1 \\ &= 2110.7 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 121 \cdot 184 \cdot 211 \\ &= 47121.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 121 \cdot 746 (314 + 37.3) \\ &= 316851.7 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 121 \cdot 746 \cdot 166 \\ &= 149861.1 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (316852 + 149861 + 47121) / 121 \\ &= 4241.5 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2111 / (2 \cdot 4241 / 362 - 746 / 86.5) \\ &= 142.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1407 [64.8 / 746] \\ &= 122.3 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:
= min(Pmax1, Pmax2)
= min(143, 122)
= 122.341 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (47121 + 316852 + 149861) / 362
= 1418.357 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 121 • 746 / 86.5
= 1044.0 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 1418 - 1044, 1044)
= 1792.7 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1792.68	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	122.34	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
= (Rnc + tn) / Rnc
= (166 + 37.3) / 166
= 1.225 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [LT]:
= $\pi/2 \cdot (Rn + tn)$
= $\pi/2 \cdot (166 + 37.3)$
= 319.186 mm

Weld Length of Pad to Shell Weld [LTP]:
= $\pi/2 \cdot (Rn + tn + W)$
= $\pi/2 \cdot (166 + 37.3 + 100)$
= 476.266 mm

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 48.083, 0.000, mm

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 37.287, 6.000)$ Passed
tf1, L42T = 48.083, must be $\geq \min(0.6 \cdot 68.000, 0.6 \cdot 64.825)$ Passed

Weld Load Value [fwelds]:
= min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
= min(149861 • 1.22, 1.5 • 1203(78.9+0), $\pi/4 \cdot 121 \cdot 166^2 \cdot 1.22^2$)
= 39286.438 kgf

Discontinuity Force [fws]:
= fwelds • t • S / (t • S + te • Sp)
= 39286 • 64.8 • 1407 / (64.8 • 1407 + 68 • 1407)
= 19173.637 kgf

Discontinuity Force [fwp]:
= fwelds • te • Sp / (t • S + te • Sp)
= 39286 • 68 • 1407 / (64.8 • 1407 + 68 • 1407)
= 20112.799 kgf

Shear Stress [τ_1]:
= fws / (LT • (0.6 • tw1 + 0.49 • L43T))
= 19174 / (319 • (0.6 • 64.8 + 0.49 • 0))
= 154.443 kgf/cm²



Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T})) \\ &= 20113 / (319 \cdot (0.6 \cdot 68 + 0.49 \cdot 7.07)) \\ &= 142.354 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T})) \\ &= 20113 / (476 \cdot (0.49 \cdot 48.1)) \\ &= 179.241 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(154, 142, 179) \\ &= 179.2 \text{ must be less than or equal to } 1407.1 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -6 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 24 °C
Temperature reduction per Graph 3.13	: 30 °C
The computed stress ratio is	: 0.545

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (17.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.545 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -27 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 30 °C
The computed stress ratio is	: 0.545

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (17.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.545 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -27 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 30 °C
The computed stress ratio is	: 0.545

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (17.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.545 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Shell to Pad Weld Junction at Pad OD, Curve: B



This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 13 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.13 : 1 °C
 The computed stress ratio is : 0.975

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (63.2 \cdot 1) / (68 - 3.17 - 0)$$

$$= 0.975$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -27 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 3 °C
 Temperature reduction per Graph 3.13 : 30 °C
 The computed stress ratio is : 0.545

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (17.6 \cdot 1) / (35.4 - 3.17 - 0)$$

$$= 0.545$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the Nozzle : -6 °C
 Gov. MDMT of the Reinforcement Pad : 13 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 13 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -30 °C

The Drop for this Nozzle is : 30.1420 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 398.1420 mm

Input Echo, WRC297 Item 1, Description: N1-Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1400.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	325.476	mm
Nozzle Thickness	40.4622	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad : 606.400 mm
 Thickness of Reinforcing Pad : 68.0000 mm

**Note:**

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	121.15	kgf/cm ²
Radial Load	P	-4283.03	kgf
Circumferential Shear	Vc	3263.25	kgf
Longitudinal Shear	Vl	3263.25	kgf
Circumferential Moment	Mc	4077.42	kgf-m.
Longitudinal Moment	Ml	4077.42	kgf-m.
Torsional Moment	Mt	4892.90	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(769.588(68.0 - 3.175))}$$

$$= 1143.479 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1539.175 mm
Nozzle Outside Diameter	(d) =	406.400 mm
Vessel Thickness used	(T) =	132.825 mm
Nozzle Thickness used	(t) =	37.287 mm
T / t	=	3.562
d / t	=	10.899

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.899$$

Nr/P	=	0.162
Mr/P	=	0.120
M0/P	=	0.170
N0/P	=	0.212
MrD/Mc	=	0.161
NrDL/Mc	=	0.171
M0d/Mc	=	0.229
N0DL/Mc	=	0.074
MrD/Ml	=	0.140
NrDL/Ml	=	0.154
M0D/Ml	=	0.208
N0DL/Ml	=	0.155

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-71	71	71	-71
Normal Pressure Stress	585	707	585	707
Outplane Stress Summary	536	749	695	625

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	17	-17	17	-17



Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	292	292	292	292

Inplane Stress Summary	257	318	370	240
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	14	14	14	14

Shear Stress Summary	17	17	10	10
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	537	750	695	625
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	292	292	292	292

Outplane Stress Summary	240	347	405	199
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-54	54	54	-54
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	585	707	585	707

Inplane Stress Summary	542	738	671	648
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	14	14	14	14

Shear Stress Summary	10	10	17	17
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	542	738	672	649

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	581	726	581	726
Hoop Stress Summary	577	723	595	740

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	71	-71	71	-71
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-111	-111	111	111
Axial Bending (ML)	-270	270	270	-270
Axial Pressure Stress	290	290	290	290
Axial Stress Summary	-9	387	753	70

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	13	13	-13	-13
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	50	50	50	50
Shear Stress Summary	64	64	36	36

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	601	734	761	742

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	581	726	581	726
Hoop Stress Summary	582	727	590	736

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	9	9	9	9
Axial Bending (P)	71	-71	71	-71
Axial Membrane (Mc)	-111	-111	111	111
Axial Bending (Mc)	-328	328	328	-328
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	290	290	290	290

Axial Stress Summary	-67	445	811	12
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-13	-13	13	13
Shear due to Torsion	50	50	50	50
Shear Stress Summary	36	36	64	64

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	654	732	828	741

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(735.588(68.0 - 3.175))}$
 $= 1127.013 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1471.175 mm
Nozzle Outside Diameter (d)	=	606.400 mm
Vessel Thickness used (T)	=	64.825 mm
Nozzle Thickness used (t)	=	137.287 mm
T / t	=	0.472
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.964$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.069
Mr/P	=	0.144
M0/P	=	0.042
N0/P	=	0.097
MrD/Mc	=	0.285
NrDL/Mc	=	0.099
M0d/Mc	=	0.088
N0DL/Mc	=	0.070
MrD/Ml	=	0.113
NrDL/Ml	=	0.068
M0D/Ml	=	0.037
N0DL/Ml	=	0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19



Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	1256	1377	1256	1377
Outplane Stress Summary	1236	1377	1346	1345

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	87	-87	87	-87
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-108	108	108	-108
Inplane Pressure Stress	628	628	628	628
Inplane Stress Summary	603	645	842	449

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	13	13	13	13
Shear Stress Summary	18	18	7	7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1236	1378	1346	1345

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-84	84	84	-84
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	628	628	628	628
Outplane Stress Summary	567	686	759	538

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	87	-87	87	-87
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-273	273	273	-273
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1256	1377	1256	1377
Inplane Stress Summary	1061	1554	1640	1038

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----

(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-5	-5	5	5
Torsional Shear (Mt)			13	13	13	13
Shear Stress Summary			7	7	18	18

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		1061	1554	1640	1039

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	585	707	585	707	585	707	585	707
Circ. Pl (SUS)	-3	-3	13	13	-6	-6	12	12
Circ. Q (SUS)	-47	47	95	-95	-37	37	71	-71
Long. Pm (SUS)	292	292	292	292	292	292	292	292
Long. Pl (SUS)	-5	-5	11	11	1	1	9	9
Long. Q (SUS)	-30	30	64	-64	-54	54	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	585.0	707.0	585.0	707.0	585.0	707.0	585.0	707.0
Pm+Pl (SUS)	582.0	704.0	598.0	720.0	579.0	701.0	597.0	719.0
Pm+Pl+Q (Total)	536.0	751.7	693.4	625.3	542.4	738.3	669.1	648.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	707.00	1407.13	Passed
Pm+Pl (SUS)	720.02	2110.69	Passed
Pm+Pl+Q (TOTAL)	751.66	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1256	1377	1256	1377	1256	1377	1256	1377
Circ. Pl (SUS)	-10	-10	28	28	-8	-8	22	22
Circ. Q (SUS)	-10	10	60	-60	-186	186	360	-360
Long. Pm (SUS)	628	628	628	628	628	628	628	628
Long. Pl (SUS)	-3	-3	17	17	-2	-2	20	20
Long. Q (SUS)	-21	21	195	-195	-59	59	109	-109

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	1256	1377	1256	1377	1256	1377	1256	1377
Pm+Pl (SUS)	1246	1367	1284	1405	1248	1369	1278	1399
Pm+Pl+Q (Total)	1236	1377	1344	1345	1062	1555	1638	1039

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1377.00	1407.13	Passed
Pm+Pl (SUS)	1405.03	2110.69	Passed
Pm+Pl+Q (TOTAL)	1638.37	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	581	726	581	726	581	726	581	726
Circ. Pl (SUS)	-3	-3	13	13	1	1	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	290	290	290	290	290	290	290	290
Long. Pl (SUS)	-102	-102	120	120	-102	-102	120	120
Long. Q (SUS)	-199	199	341	-341	-257	257	399	-399
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	13	13	-13	-13	-13	-13	13	13
Shear Q (SUS)	50	50	50	50	50	50	50	50
Pm (SUS)	581.0	726.0	581.0	726.0	581.0	726.0	581.0	726.0
Pm+Pl (SUS)	578.4	723.3	594.9	739.5	582.4	727.3	590.9	735.5
Pm+Pl+Q (Total)	602.3	734.4	759.3	741.0	655.2	731.8	825.8	740.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	726.00	1203.20	Passed
Pm+Pl (SUS)	739.51	1804.80	Passed
Pm+Pl+Q (TOTAL)	825.83	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N2-Tube Outlet

From: 30

Pressure for Reinforcement Calculations	P	121.284	kgf/cm ²
Temperature for Internal Pressure	Temp	66	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1400.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		146.32	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

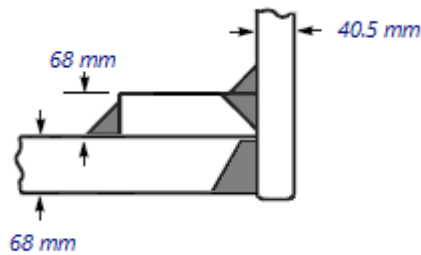
Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	68.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	606.4000	mm
Thickness of Pad	te	68.0000	mm
Weld leg size between Pad and Shell	Wp	68.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	68.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 332 \cdot (\exp(121/(1203 \cdot 1)) - 1) + 3.17 + 0$$

$$= 17.6 + 3.17 + 0 = 20.8 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1203 \cdot 1 \cdot \ln((2 \cdot 37.3 + 332) / 332) - 0$$

$$= 244 - 0 = 244 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 251.80 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 516.70 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm
 Min. required shell thickness, internal p + c : 66.513 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 35.408 mm , is greater than required thickness 11.515 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 458.6,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2549.4 kgf/cm ²	Passed
Occasional	: 284.5,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 115.2,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1203/1407, 1]$$

$$= 0.855$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1407/1407, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:



= thickness - corrosion allowance
= 40.5 - 3.17
= 37.287 mm

Shell Diameter to Thickness ratio [D/t]:
= Di/t
= 1406/64.8
= 21.695 must be less than 400.

Effective Pressure Radius [Reff]:
= Di/2 + corrosion allowance
= 1400/2 + 3.17
= 703.175 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:
= $\sqrt{(Reff \cdot t) + W}$
= $\sqrt{(703 \cdot 64.8) + 100}$
= 313.503 mm

Limit [LR2]:
= $\sqrt{((Reff + t) \cdot (t + te))}$
= $\sqrt{((703 + 64.8) \cdot (64.8 + 68))}$
= 319.389 mm

Limit [LR3]:
= 2 • Rn
= 2 • 166
= 331.826 mm

Final value of [LR]:
= min(LR1, LR2, LR3)
= min(314, 319, 332)
= 313.503 mm

Thickness Limit Candidate [LH1]:
= min[1.5t, te] + $\sqrt{(Rn \cdot tn)}$
= min[1.5•64.8, 68] + $\sqrt{(166 \cdot 37.3)}$
= 146.654 mm

Thickness Limit Candidate [LH2]:
= Lpr1
= 300
= 300.000 mm

Thickness Limit Candidate [LH3]:
= 8(t + te)
= 8(64.8 + 68)
= 1062.600 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
= min[LH1, LH2, LH3] + t for set-in nozzles
= min[147, 300, 1063] + 64.8
= 211.479 mm

Effective Vessel Thickness [teff]:
= t + ((A5 • frp) / LR)
= 64.8 + ((68•1)/314)
= 86.515 mm

Parameter [λ]:
= min(12, (2•Rn + tn) / $\sqrt{((Di + teff) \cdot teff)}$))
= min(12, (2•166 + 37.3) / $\sqrt{((1406 + 86.5) \cdot 86.5)}$))
= 1.027

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :



Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\frac{\lambda}{5}\right)^{0.85}, 1\right) \\ &= 64.8 \cdot 314 \cdot \max\left(\left(\frac{1.03}{5}\right)^{0.85}, 1\right) \\ &= 203.228 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 37.3 \cdot 211 \\ &= 78.855 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 68^2 \\ &= 23.120 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(100 \cdot 68, 314 \cdot 68) \\ &= 68.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 203 + 0.86(78.9 + 0) + 0.5 + 23.1 + 0 + 1(68) \\ &= 362.275 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 37.3 / \ln[1 + 37.3/166] \\ &= 183.927 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 86.5 / \ln[1 + 86.5/703] \\ &= 745.596 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1407 \cdot 1 \\ &= 2110.7 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 121 \cdot 184 \cdot 211 \\ &= 47175.4 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 121 \cdot 746 (314 + 37.3) \\ &= 317215.0 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 121 \cdot 746 \cdot 166 \\ &= 150033.0 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (317215 + 150033 + 47175) / 121 \\ &= 4241.5 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2111 / (2 \cdot 4241 / 362 - 746 / 86.5) \\ &= 142.6 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1407 [64.8/746] \\ &= 122.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(143, 122) \\ &= 122.341 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (47175 + 317215 + 150033) / 362 \\ &= 1419.984 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 121 \cdot 746 / 86.5 \\ &= 1045.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1420 - 1045, 1045) \\ &= 1794.7 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1794.73	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	122.34	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (166 + 37.3) / 166 \\ &= 1.225 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (166 + 37.3) \\ &= 319.186 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (166 + 37.3 + 100) \\ &= 476.266 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 48.083, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 37.287, 6.000) \text{ Passed} \\ &tf1, L42T = 48.083, \text{ must be } \geq \min(0.6 \cdot 68.000, 0.6 \cdot 64.825) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(150033 \cdot 1.22, 1.5 \cdot 1203(78.9+0), \pi/4 \cdot 121 \cdot 166^2 \cdot 1.22^2) \\ &= 39331.488 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 39331 \cdot 64.8 \cdot 1407 / (64.8 \cdot 1407 + 68 \cdot 1407) \\ &= 19195.625 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 39331 \cdot 68 \cdot 1407 / (64.8 \cdot 1407 + 68 \cdot 1407) \\ &= 20135.861 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:



$$\begin{aligned} &= fws / (Lt \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 19196 / (319 \cdot (0.6 \cdot 64.8 + 0.49 \cdot 0)) \\ &= 154.620 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (Lt \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 20136 / (319 \cdot (0.6 \cdot 68 + 0.49 \cdot 7.07)) \\ &= 142.518 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$\begin{aligned} &= fwp / (LtP \cdot (0.49 \cdot L42T)) \\ &= 20136 / (476 \cdot (0.49 \cdot 48.1)) \\ &= 179.447 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(155, 143, 179) \\ &= 179.4 \text{ must be less than or equal to } 1407.1 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -6 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 24 °C
Temperature reduction per Graph 3.13	: 30 °C
The computed stress ratio is	: 0.546

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (17.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.546 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -27 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 30 °C
The computed stress ratio is	: 0.546

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (17.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.546 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -27 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 30 °C
The computed stress ratio is	: 0.546

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (17.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.546 \end{aligned}$$



Warning: Impact Testing Is Mandatory!

Shell to Pad Weld Junction at Pad OD, Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : 13 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 14 °C
 Temperature reduction per Graph 3.13 : 1 °C
 The computed stress ratio is : 0.976

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (63.3 \cdot 1) / (68 - 3.17 - 0)$$

$$= 0.976$$

Warning: Impact Testing Is Mandatory!

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -27 °C
 User entered Minimum Design Temperature : -29 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 3 °C
 Temperature reduction per Graph 3.13 : 30 °C
 The computed stress ratio is : 0.546

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (17.6 \cdot 1) / (35.4 - 3.17 - 0)$$

$$= 0.546$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the Nozzle : -6 °C
 Gov. MDMT of the Reinforcement Pad : 13 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 13 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -30 °C

The Drop for this Nozzle is : 30.1420 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 398.1420 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1400.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	325.476	mm
Nozzle Thickness	40.4622	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter of Reinforcing Pad 606.400 mm
 Thickness of Reinforcing Pad 68.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	121.28	kgf/cm ²
Radial Load	P	-4283.02	kgf
Circumferential Shear	Vc	3263.25	kgf
Longitudinal Shear	Vl	3263.25	kgf
Circumferential Moment	Mc	4077.42	kgf-m.
Longitudinal Moment	Ml	4077.42	kgf-m.
Torsional Moment	Mt	4892.90	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(769.588(68.0 - 3.175))}$
 $= 1143.479 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1539.175 mm
Nozzle Outside Diameter	(d) =	406.400 mm
Vessel Thickness used	(T) =	132.825 mm
Nozzle Thickness used	(t) =	37.287 mm
T / t	=	3.562
d / t	=	10.899

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.899$$

Nr/P	=	0.162
Mr/P	=	0.120
M0/P	=	0.170
N0/P	=	0.212
MrD/Mc	=	0.161
NrDL/Mc	=	0.171
M0d/Mc	=	0.229
N0DL/Mc	=	0.074
MrD/Ml	=	0.140
NrDL/Ml	=	0.154
M0D/Ml	=	0.208
N0DL/Ml	=	0.155

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-8	-8	8	8
Outplane Bending (Ml)	-71	71	71	-71
Normal Pressure Stress	586	707	586	707
Outplane Stress Summary	536	750	696	626

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	293	293	293	293
Inplane Stress Summary	258	318	370	240

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	14	14	14	14
Shear Stress Summary	17	17	10	10

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	537	751	696	626

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	24	-24	24	-24
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	293	293	293	293
Outplane Stress Summary	240	347	405	199

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-54	54	54	-54
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	586	707	586	707
Inplane Stress Summary	543	739	672	649

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	14	14	14	14
Shear Stress Summary	10	10	17	17



Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	543	739	673	649

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	582	727	582	727
Hoop Stress Summary	578	724	596	741

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	71	-71	71	-71
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-111	-111	111	111
Axial Bending (ML)	-270	270	270	-270
Axial Pressure Stress	291	291	291	291
Axial Stress Summary	-9	388	753	70

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	13	13	-13	-13
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	50	50	50	50
Shear Stress Summary	64	64	36	36

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	601	735	762	743

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	582	727	582	727
Hoop Stress Summary	583	728	591	736

Nozzle Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	9	9	9	9
Axial Bending (P)	71	-71	71	-71
Axial Membrane (Mc)	-111	-111	111	111
Axial Bending (Mc)	-328	328	328	-328
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	291	291	291	291
Axial Stress Summary	-67	446	811	12

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-13	-13	13	13
Shear due to Torsion	50	50	50	50
Shear Stress Summary	36	36	64	64

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	654	733	829	742

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(735.588(68.0 - 3.175))}$$

$$= 1127.013 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1471.175 mm
Nozzle Outside Diameter	(d) =	606.400 mm
Vessel Thickness used	(T) =	64.825 mm
Nozzle Thickness used	(t) =	137.287 mm
T / t	=	0.472
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.964$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.069
Mr/P	=	0.144
M0/P	=	0.042
N0/P	=	0.097
MrD/Mc	=	0.285
NrDL/Mc	=	0.099
M0d/Mc	=	0.088
N0DL/Mc	=	0.070
MrD/Ml	=	0.113
NrDL/Ml	=	0.068
M0D/Ml	=	0.037
N0DL/Ml	=	0.122

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9



Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19
Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	1257	1378	1257	1378

Outplane Stress Summary	1237	1379	1347	1347
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	87	-87	87	-87
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-108	108	108	-108
Inplane Pressure Stress	628	628	628	628

Inplane Stress Summary	604	645	843	449
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	13	13	13	13

Shear Stress Summary	18	18	7	7
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1238	1379	1347	1347
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-84	84	84	-84
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	628	628	628	628

Outplane Stress Summary	568	686	759	539
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	87	-87	87	-87
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-273	273	273	-273
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1257	1378	1257	1378



Inplane Stress Summary 1063 1555 1641 1040

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-5	-5	5	5
Torsional Shear (Mt)		13	13	13	13

Shear Stress Summary 7 7 18 18

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		1063	1555	1642	1040

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	586	707	586	707	586	707	586	707
Circ. Pl (SUS)	-3	-3	13	13	-6	-6	12	12
Circ. Q (SUS)	-47	47	95	-95	-37	37	71	-71
Long. Pm (SUS)	293	293	293	293	293	293	293	293
Long. Pl (SUS)	-5	-5	11	11	1	1	9	9
Long. Q (SUS)	-30	30	64	-64	-54	54	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	586.0	707.0	586.0	707.0	586.0	707.0	586.0	707.0
Pm+Pl (SUS)	583.0	704.0	599.0	720.0	580.0	701.0	598.0	719.0
Pm+Pl+Q (Total)	537.0	751.7	694.4	625.3	543.4	738.3	670.1	648.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	707.00	1407.13	Passed
Pm+Pl (SUS)	720.02	2110.69	Passed
Pm+Pl+Q (TOTAL)	751.67	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1257	1378	1257	1378	1257	1378	1257	1378
Circ. Pl (SUS)	-10	-10	28	28	-8	-8	22	22
Circ. Q (SUS)	-10	10	60	-60	-186	186	360	-360

Long. Pm (SUS)	628	628	628	628	628	628	628	628
Long. Pl (SUS)	-3	-3	17	17	-2	-2	20	20
Long. Q (SUS)	-21	21	195	-195	-59	59	109	-109
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	1257	1378	1257	1378	1257	1378	1257	1378
Pm+Pl (SUS)	1247	1368	1285	1406	1249	1370	1279	1400
Pm+Pl+Q (Total)	1237	1378	1345	1346	1063	1556	1639	1040

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1378.00	1407.13	Passed
Pm+Pl (SUS)	1406.03	2110.69	Passed
Pm+Pl+Q (TOTAL)	1639.37	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	582	727	582	727	582	727	582	727
Circ. Pl (SUS)	-3	-3	13	13	1	1	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	291	291	291	291	291	291	291	291
Long. Pl (SUS)	-102	-102	120	120	-102	-102	120	120
Long. Q (SUS)	-199	199	341	-341	-257	257	399	-399
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	13	13	-13	-13	-13	-13	13	13
Shear Q (SUS)	50	50	50	50	50	50	50	50
Pm (SUS)	582.0	727.0	582.0	727.0	582.0	727.0	582.0	727.0
Pm+Pl (SUS)	579.4	724.3	595.9	740.5	583.4	728.3	591.9	736.5
Pm+Pl+Q (Total)	602.3	735.4	760.3	742.0	655.2	732.8	826.8	741.4

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	727.00	1203.20	Passed
Pm+Pl (SUS)	740.51	1804.80	Passed
Pm+Pl+Q (TOTAL)	826.83	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N3-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	157.628	kgf/cm ²
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1400.00	mm
Shell Finished (Minimum) Thickness	t	88.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		323.32	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

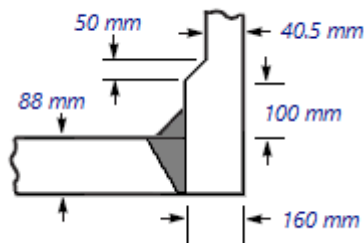
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.13	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.13	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-266 2	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	88.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 332 \cdot (\exp(158/(1407 \cdot 1)) - 1) + 3.17 + 0$
 $= 19.7 + 3.17 + 0 = 22.8 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 37.3 + 332) / 332) - 0$
 $= 285 - 0 = 285 \text{ kgf/cm}^2$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 300.46 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 566.96 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm
 Min. required shell thickness, internal p + c : 86.619 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 35.408 mm , is greater than required thickness 11.515 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 578.7,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2939.1 kgf/cm ²	Passed
Occasional	: 369.8,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 136.7,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1407/1407, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 160 - 3.17$
 $= 156.825 \text{ mm}$

Thickness of Nozzle at Top [tn2]:



= thickness - corrosion allowance
= 40.5 - 3.17
= 37.287 mm

Shell Diameter to Thickness ratio [D/t]:
= Di/t
= 1406/84.8
= 16.579 must be less than 400.

Effective Pressure Radius [Reff]:
= Di/2 + corrosion allowance
= 1400/2 + 3.17
= 703.175 mm

Effective Length of Vessel Wall [LR]:
= $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
= $\min(\sqrt{703 \cdot 84.8}, 2 \cdot 166)$
= 244.227 mm

Thickness Limit Candidate [LH1]:
= $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
= $\min[1.5 \cdot 84.8, 0] + \sqrt{166 \cdot 157}$
= 161.305 mm

Thickness Limit Candidate [LH2]:
= Lpr1
= 300
= 300.000 mm

Thickness Limit Candidate [LH3]:
= $8(t + te)$
= $8(84.8 + 0)$
= 678.600 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[161, 300, 679] + 84.8$
= 246.130 mm

Effective Vessel Thickness [teff]:
= $t + ((A5 \cdot frp) / LR)$
= $84.8 + ((0 \cdot 1) / 244)$
= 84.825 mm

Parameter [λ]:
= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 166 + 157) / (\sqrt{(1406 + 84.8) \cdot 84.8}))$
= 1.374

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $84.8 \cdot 244 \cdot \max((1.37/5)^{0.85}, 1)$
= 207.166 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
= A2A + A2C
= 290 + 22.9
= 312.728 cm²

Area Contributed by the Outside Fillet Weld [A41]:
= $0.5 \cdot Leg^2$
= $0.5 \cdot 10^2$
= 0.500 cm²

The total area contributed by A1 through A43 [AT]:
= $A1 + \min(A2 + A3) + A41 + A42 + A43$
= $207 + 1(313 + 0) + 0.5 + 0 + 0$
= 520.393 cm²



Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 157 / \ln[1 + 157/166] \\ &= 235.693 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 84.8 / \ln[1 + 84.8/703] \\ &= 744.782 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1407 \cdot 1 \\ &= 2110.7 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 158 \cdot 236 \cdot 246 \\ &= 91441.4 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 158 \cdot 745 (244 + 157) \\ &= 470827.3 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 158 \cdot 745 \cdot 166 \\ &= 194778.4 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (470827 + 194778 + 91441) / 158 \\ &= 4802.8 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2111 / (2 \cdot 4803 / 520 - 745 / 84.8) \\ &= 218.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1407 [84.8 / 745] \\ &= 160.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(218, 160) \\ &= 160.261 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (91441 + 470827 + 194778) / 520 \\ &= 1454.762 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 158 \cdot 745 / 84.8 \\ &= 1384.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1455 - 1384, 1384) \\ &= 1525.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1525.52	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	160.26	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (166 + 157) / 166 \\ &= 1.945 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (166 + 157) \\ &= 506.955 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 156.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(194778 \cdot 1.95, 1.5 \cdot 1407(313+0), \pi/4 \cdot 158 \cdot 166^2 \cdot 1.95^2) \\ &= 128949.938 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 128950 / (507(0.49 \cdot 7.07 + 0.6 \cdot 84.8 + 0.49 \cdot 0)) \\ &= 467.923 < \text{or} = \text{to } 1407.129 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -20 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 23 °C
The computed stress ratio is	: 0.610

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (19.7 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.610 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -20 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -40 °C

The Drop for this Nozzle is : 78.8395 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 466.8395 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1400.000	mm
Shell Thickness	88.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	



Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.1750	mm
Nozzle Diameter		325.476	mm
Nozzle Thickness		160.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	1407.13	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	157.63	kgf/cm ²
Radial Load	P	-5098.83	kgf
Circumferential Shear	Vc	4079.07	kgf
Longitudinal Shear	Vl	4079.07	kgf
Circumferential Moment	Mc	4892.90	kgf-m.
Longitudinal Moment	Ml	4892.90	kgf-m.
Torsional Moment	Mt	5708.38	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, D_v/T_v (17.909) is < 20
check the limitations of WRC bulletin 297.*

*Warning - The ratio, D_n/T_n (4.034) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 645.476 + 2 \cdot 1.65 \cdot \sqrt{(745.588(88.0 - 3.175))}$$

$$= 1475.375 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1491.175 mm
Nozzle Outside Diameter	(d) =	645.476 mm
Vessel Thickness used	(T) =	84.825 mm
Nozzle Thickness used	(t) =	156.825 mm
T / t	=	0.541
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.815$$

Nr/P	=	0.074
Mr/P	=	0.149
M0/P	=	0.044
N0/P	=	0.102
MrD/Mc	=	0.285
NrDL/Mc	=	0.103
M0d/Mc	=	0.090
N0DL/Mc	=	0.065
MrD/Ml	=	0.122
NrDL/Ml	=	0.070
M0D/Ml	=	0.040
N0DL/Ml	=	0.128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	1232	1389	1232	1389
Outplane Stress Summary	1219	1390	1296	1366

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	616	616	616	616
Inplane Stress Summary	600	627	769	488

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	14	14	5	5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1219	1390	1296	1366

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	616	616	616	616
Outplane Stress Summary	578	654	705	554

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-180	180	180	-180
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1232	1389	1232	1389
Inplane Stress Summary	1109	1501	1491	1162

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	10	10	10	10
Shear Stress Summary	5	5	14	14

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1109	1501	1492	1162

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	135	325	135	325
Hoop Stress Summary	129	318	156	345

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-16	16	16	-16
Axial Pressure Stress	67	67	67	67
Axial Stress Summary	48	52	119	59

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5
Shear Stress Summary	8	8	3	3

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	130	319	156	345

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	135	325	135	325
Hoop Stress Summary	136	325	149	339

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-43	43	43	-43
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	67	67	67	67
Axial Stress Summary	20	79	147	32

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	5	5	5	5
Shear Stress Summary	3	3	8	8

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	136	325	156	339

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1232	1389	1232	1389	1232	1389	1232	1389
Circ. Pl (SUS)	-6	-6	20	20	-5	-5	15	15
Circ. Q (SUS)	-7	7	43	-43	-117	117	243	-243
Long. Pm (SUS)	616	616	616	616	616	616	616	616
Long. Pl (SUS)	-2	-2	12	12	1	1	13	13
Long. Q (SUS)	-14	14	140	-140	-38	38	74	-74
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	1232	1389	1232	1389	1232	1389	1232	1389
Pm+Pl (SUS)	1226	1383	1252	1409	1227	1384	1247	1404
Pm+Pl+Q (Total)	1219	1390	1295	1366	1110	1501	1490	1161



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1389.00	1407.13	Passed
Pm+Pl (SUS)	1409.02	2110.69	Passed
Pm+Pl+Q (TOTAL)	1501.04	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	325	135	325	135	325	135	325
Circ. Pl (SUS)	-6	-6	20	20	1	1	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-17	-17	21	21	-17	-17	21	21
Long. Q (SUS)	-3	3	29	-29	-30	30	56	-56
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	135.0	325.0	135.0	325.0	135.0	325.0	135.0	325.0
Pm+Pl (SUS)	129.1	319.0	155.1	345.0	136.0	326.0	148.1	338.0
Pm+Pl+Q (Total)	129.6	319.2	155.2	345.0	136.1	326.0	153.3	338.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	325.00	1407.13	Passed
Pm+Pl (SUS)	345.02	2110.69	Passed
Pm+Pl+Q (TOTAL)	345.03	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4-Shell Outlet

From: 40

Pressure for Reinforcement Calculations	P	157.489	kgf/cm ²
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1400.00	mm
Shell Finished (Minimum) Thickness	t	88.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		753.32	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

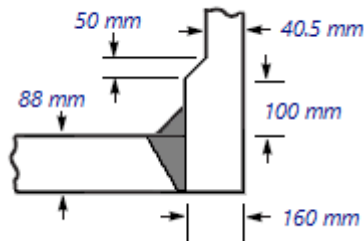
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1407.14	kgf/cm ²
Allowable Stress At Ambient	Sna	1407.14	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-266 2	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm
Corrosion Allowance	can	3.1750	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	88.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N4-Shell Outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	325.476	mm.
Actual Thickness Used in Calculation	40.462	mm



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 332 \cdot (\exp(157/(1407 \cdot 1)) - 1) + 3.17 + 0$
 $= 19.6 + 3.17 + 0 = 22.8 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1407 \cdot 1 \cdot \ln((2 \cdot 37.3 + 332) / 332) - 0$
 $= 285 - 0 = 285 \text{ kgf/cm}^2$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 552.26 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 566.96 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.515 mm
 Min. required shell thickness, internal p + c : 86.541 mm

Nozzle neck minimum required thickness : 11.515 mm

Minimum nozzle neck thickness 35.408 mm , is greater than required thickness 11.515 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 578.4,	Allowable	: 1407.1 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2939.4 kgf/cm ²	Passed
Occasional	: 369.5,	Allowable	: 1871.5 kgf/cm ²	Passed
Shear	: 136.7,	Allowable	: 985.0 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1407/1407, 1]$
 $= 1.000$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 160 - 3.17$
 $= 156.825 \text{ mm}$

Thickness of Nozzle at Top [tn2]:



= thickness - corrosion allowance
 = 40.5 - 3.17
 = 37.287 mm

Shell Diameter to Thickness ratio [D/t]:
 = Di/t
 = 1406/84.8
 = 16.579 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 1400/2 + 3.17
 = 703.175 mm

Effective Length of Vessel Wall [LR]:
 = $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
 = $\min(\sqrt{703 \cdot 84.8}, 2 \cdot 166)$
 = 244.227 mm

Thickness Limit Candidate [LH1]:
 = $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
 = $\min[1.5 \cdot 84.8, 0] + \sqrt{166 \cdot 157}$
 = 161.305 mm

Thickness Limit Candidate [LH2]:
 = Lpr1
 = 300
 = 300.000 mm

Thickness Limit Candidate [LH3]:
 = $8(t + te)$
 = $8(84.8 + 0)$
 = 678.600 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:
 = $\min[LH1, LH2, LH3] + t$ for set-in nozzles
 = $\min[161, 300, 679] + 84.8$
 = 246.130 mm

Effective Vessel Thickness [teff]:
 = $t + ((A5 \cdot frp) / LR)$
 = $84.8 + ((0 \cdot 1) / 244)$
 = 84.825 mm

Parameter [λ]:
 = $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
 = $\min(12, (2 \cdot 166 + 157) / (\sqrt{(1406 + 84.8) \cdot 84.8}))$
 = 1.374

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
 = $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
 = $84.8 \cdot 244 \cdot \max((1.37/5)^{0.85}, 1)$
 = 207.166 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
 = A2A + A2C
 = 290 + 22.9
 = 312.728 cm²

Area Contributed by the Outside Fillet Weld [A41]:
 = $0.5 \cdot Leg^2$
 = $0.5 \cdot 10^2$
 = 0.500 cm²

The total area contributed by A1 through A43 [AT]:
 = $A1 + \min(A2 + A3) + A41 + A42 + A43$
 = $207 + 1(313 + 0) + 0.5 + 0 + 0$
 = 520.393 cm²

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 157 / \ln[1 + 157/166] \\ &= 235.693 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 84.8 / \ln[1 + 84.8/703] \\ &= 744.782 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1407 \cdot 1 \\ &= 2110.7 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 157 \cdot 236 \cdot 246 \\ &= 91360.8 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (L_R + t_n) \\ &= 157 \cdot 745 (244 + 157) \\ &= 470412.4 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 157 \cdot 745 \cdot 166 \\ &= 194606.8 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (470412 + 194607 + 91361) / 157 \\ &= 4802.8 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2111 / (2 \cdot 4803 / 520 - 745 / 84.8) \\ &= 218.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1407 [84.8 / 745] \\ &= 160.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(218, 160) \\ &= 160.261 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (91361 + 470412 + 194607) / 520 \\ &= 1453.481 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 157 \cdot 745 / 84.8 \\ &= 1382.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1453 - 1383, 1383) \\ &= 1524.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2110.69	kgf/cm ²
Local Primary Membrane Stress	PL	1524.18	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	160.26	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (166 + 157) / 166 \\ &= 1.945 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (166 + 157) \\ &= 506.955 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 156.825, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(194607 \cdot 1.95, 1.5 \cdot 1407(313+0), \pi/4 \cdot 157 \cdot 166^2 \cdot 1.95^2) \\ &= 128836.305 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 128836 / (507(0.49 \cdot 7.07 + 0.6 \cdot 84.8 + 0.49 \cdot 0)) \\ &= 467.510 < \text{or } = \text{ to } 1407.129 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: -21 °C
User entered Minimum Design Temperature	: -29 °C
The Curve for this Material is	: B
From Figure 3.8M- with PWHT, the MDMT is	: 3 °C
Temperature reduction per Graph 3.13	: 23 °C
The computed stress ratio is	: 0.610

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (19.6 \cdot 1) / (35.4 - 3.17 - 0) \\ &= 0.610 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -40 °C

The Drop for this Nozzle is : 78.8395 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 466.8395 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1400.000	mm
Shell Thickness	88.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	



Vessel Cold S.I. Allowable	Smc	1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.1750	mm
Nozzle Diameter		325.476	mm
Nozzle Thickness		160.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	1407.14	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1407.14	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	157.49	kgf/cm ²
Radial Load	P	-5098.83	kgf
Circumferential Shear	Vc	4079.07	kgf
Longitudinal Shear	Vl	4079.07	kgf
Circumferential Moment	Mc	4892.90	kgf-m.
Longitudinal Moment	Ml	4892.90	kgf-m.
Torsional Moment	Mt	5708.38	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (17.909) is < 20
check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (4.034) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 645.476 + 2 \cdot 1.65 \cdot \sqrt{(745.588(88.0 - 3.175))}$$

$$= 1475.375 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1491.175 mm
Nozzle Outside Diameter	(d) =	645.476 mm
Vessel Thickness used	(T) =	84.825 mm
Nozzle Thickness used	(t) =	156.825 mm
T / t	=	0.541
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.815$$

Nr/P	=	0.074
Mr/P	=	0.149
M0/P	=	0.044
N0/P	=	0.102
MrD/Mc	=	0.285
NrDL/Mc	=	0.103
M0d/Mc	=	0.090
N0DL/Mc	=	0.065
MrD/Ml	=	0.122
NrDL/Ml	=	0.070
M0D/Ml	=	0.040
N0DL/Ml	=	0.128

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	1231	1388	1231	1388
Outplane Stress Summary	1218	1389	1295	1365

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	615	615	615	615
Inplane Stress Summary	599	627	768	487

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	10	10	10	10
Shear Stress Summary	14	14	5	5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1218	1389	1295	1365

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-56	56	56	-56
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	615	615	615	615
Outplane Stress Summary	577	654	705	554

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-180	180	180	-180
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1231	1388	1231	1388
Inplane Stress Summary	1108	1499	1490	1161

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	10	10	10	10
Shear Stress Summary	5	5	14	14

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1108	1499	1491	1161

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	135	324	135	324
Hoop Stress Summary	129	318	156	345

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-16	16	16	-16
Axial Pressure Stress	67	67	67	67
Axial Stress Summary	47	52	119	59

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	5	5	5	5
Shear Stress Summary	8	8	3	3

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	130	318	156	345

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	135	324	135	324
Hoop Stress Summary	136	325	149	338

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-19	-19	19	19
Axial Bending (Mc)	-43	43	43	-43
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	67	67	67	67
Axial Stress Summary	20	79	146	32

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	5	5	5	5
Shear Stress Summary	3	3	8	8

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	136	325	156	338

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1231	1388	1231	1388	1231	1388	1231	1388
Circ. Pl (SUS)	-6	-6	20	20	-5	-5	15	15
Circ. Q (SUS)	-7	7	43	-43	-117	117	243	-243
Long. Pm (SUS)	615	615	615	615	615	615	615	615
Long. Pl (SUS)	-2	-2	12	12	1	1	13	13
Long. Q (SUS)	-14	14	140	-140	-38	38	74	-74
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	1231	1388	1231	1388	1231	1388	1231	1388
Pm+Pl (SUS)	1225	1382	1251	1408	1226	1383	1246	1403
Pm+Pl+Q (Total)	1218	1389	1294	1365	1109	1500	1489	1160



Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1388.00	1407.13	Passed
Pm+Pl (SUS)	1408.02	2110.69	Passed
Pm+Pl+Q (TOTAL)	1500.04	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	324	135	324	135	324	135	324
Circ. Pl (SUS)	-6	-6	20	20	1	1	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-17	-17	21	21	-17	-17	21	21
Long. Q (SUS)	-3	3	29	-29	-30	30	56	-56
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	135.0	324.0	135.0	324.0	135.0	324.0	135.0	324.0
Pm+Pl (SUS)	129.1	318.0	155.1	344.0	136.0	325.0	148.1	337.0
Pm+Pl+Q (Total)	129.6	318.2	155.2	344.0	136.1	325.0	153.3	337.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	324.00	1407.14	Passed
Pm+Pl (SUS)	344.02	2110.71	Passed
Pm+Pl+Q (TOTAL)	344.03	4221.42	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	157.49	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	157.49	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	88.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1400.000	mm
Shell Temperature for Internal Pressure	Ts	93.33	°C
Shell Material		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1630.99	kgf/cm ²
Shell Allowable Stress at Ambient		1784.41	kgf/cm ²

Channel Data:

Channel Description: ch shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	121.14	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	121.14	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	68.0000	mm
Channel Internal Corrosion Allowance	cac	3.1750	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1400.000	mm
Channel Design Temperature	TEMPC	65.56	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1671.44	kgf/cm ²
Channel Allowable Stress at Ambient		1784.41	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	2298	
Tube Wall Thickness	et	3.0480	mm
Tube Outside Diameter	D	19.0500	mm
Total Straight Tube Length	Lt	6096.00	mm
Straight Tube Length (bet. inner tubsht faces) L		5776.00	mm
Design Temperature of the Tubes		93.33	°C
Tube Material		SA-179	
Is this a Welded Tube		No	
Tube Allowable Stress at Temperature		942.17	kgf/cm ²
Tube Allowable Stress At Ambient		942.17	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1671.78	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	3.0480	mm
Groove Weld Leg	ag	3.0480	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	681.025	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a



Tubesheet Design Metal Temperature T 93.33 °C
Tubesheet Material SA-266 2

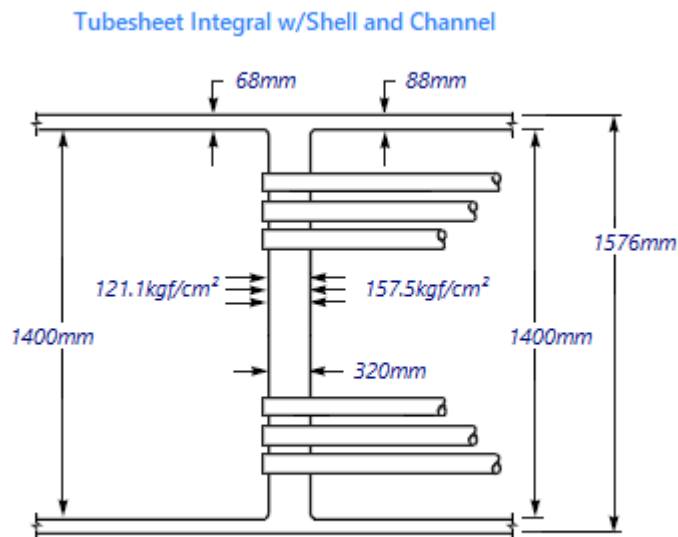
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
Thickness of Tubesheet h 320.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 3.1250 mm
Tubesheet Corr. Allowance (Channel side) Catc 3.1250 mm
Tubesheet Outside Diameter A 1576.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Tubesheet Integral with Both
Tubesheet Extended as Flange No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	93.3 °C	0.20248E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	65.6 °C	0.20411E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	93.3 °C	0.20248E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TM-1 Carbon Steels with C> 0.3%



Elastic Mod. at Design Temperature	93.3 °C	0.20108E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20530E+07 kgf/cm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 13 \cdot (\exp(121/(942 \cdot 1)) - 1) + 0$
 $= 0.89 + 0$
 $= 0.89 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 6457 / \sqrt{(9.52 \cdot 3.05)}$
 $= 1198.374$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 / (19/3.05)$
 $= 0.0880$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.088 \cdot 2024848 (3.05/19)$
 $= 45615.789 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 45615.8/1671.8$
 $= 27.286$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 1672/3304$
 $= 0.506$

Ratio [AE]:

$= Fhe / E$
 $= 45616/2024848$
 $= 0.022528$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.51) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.148$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 1672 (1 + 0.0020) / \ln(1 + 0.0020)^{0.14842063}$
 $= 4213.993 \text{ kgf/cm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f(m2f2 - R)$
 $= 0.6 (1 - 0.51)$
 $= 0.296$

Curve Fitting Constant [A2]:

$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
 $= (3304 \cdot \exp(0.3)) / (0.3^{0.3})$
 $= 6373.556 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:

$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.511.5 - 0.5 \cdot 0.512.5 - 0.513.5$
 $= 0.357$



Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1652	-2.067	0.371E-05	0.340E-05	0.107E-04	-0.928E-05	0.110E+06	0.754E-02
2	2478	0.770	0.380E-04	-0.525E-04	0.281E-04	0.594E-04	0.136E+05	0.160E+00
3	2065	-0.648	0.134E-04	-0.186E-05	0.182E-04	0.155E-04	0.219E+05	0.719E-01
4	1858	-1.358	0.730E-05	0.478E-05	0.142E-04	-0.618E-05	0.485E+05	0.158E-01
5	1755	-1.713	0.526E-05	0.435E-05	0.124E-04	-0.892E-05	0.736E+05	0.133E-02
6	1703	-1.890	0.443E-05	0.390E-05	0.116E-04	-0.929E-05	0.902E+05	0.364E-02
7	1729	-1.801	0.483E-05	0.413E-05	0.120E-04	-0.916E-05	0.815E+05	0.130E-02
8	1742	-1.757	0.504E-05	0.424E-05	0.122E-04	-0.906E-05	0.774E+05	0.246E-04
9	1749	-1.735	0.515E-05	0.430E-05	0.123E-04	-0.899E-05	0.755E+05	0.644E-03
10	1745	-1.746	0.509E-05	0.427E-05	0.122E-04	-0.902E-05	0.765E+05	0.307E-03
11	1744	-1.751	0.507E-05	0.426E-05	0.122E-04	-0.904E-05	0.769E+05	0.141E-03
12	1743	-1.754	0.505E-05	0.425E-05	0.122E-04	-0.905E-05	0.772E+05	0.579E-04
13	1742	-1.755	0.505E-05	0.425E-05	0.122E-04	-0.905E-05	0.773E+05	0.166E-04
14	1742	-1.756	0.504E-05	0.425E-05	0.122E-04	-0.905E-05	0.774E+05	0.401E-05
15	1742	-1.756	0.504E-05	0.425E-05	0.122E-04	-0.905E-05	0.773E+05	0.630E-05
16	1742	-1.756	0.504E-05	0.425E-05	0.122E-04	-0.905E-05	0.774E+05	0.115E-05

Factor of Safety [FS]:

Since Fic [1743] >= Fy[1672], FS = 1.667

Allowable Circumferential Compressive Stress [Fha]:

= Fic/FS

= 1743/1.67

= 1045.484 kgf/cm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha(t / Do)

= 2 • 1045 (3.05/19)

= 334.555 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.7513 mm
Allowable Internal Pressure at Corroded thickness	363.36 kgf/cm ²
Required Internal Design Pressure	121.14 kgf/cm ²
Allowable External Pressure at Corroded thickness	334.55 kgf/cm ²
Required External Design Pressure	157.49 kgf/cm ²
Required Thickness due to Shell Side pressure	1.7513 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= 2 • ro + dt

= 2 • 681 + 19

= 1381.100 mm

Basic ligament efficiency for shear [μ]:

= (p - dt) / p

= (23.8 - 19)/23.8

= 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

= Ds / Do (Configurations a, b, c)

= 1400/1381



$$= 1.0137$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 1400/1381 \\ &= 1.0137 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0/320 \\ &= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(d_t - 2t_t(E_t T/E) \cdot (S_t T/S) \cdot (\rho); d_t - 2t_t) \\ &= \max(19 - 2 \cdot 3.05 (2024848/2010787) \cdot \\ &\quad (942/1407) \cdot (0); 19 - 2 \cdot 3.05) \\ &= 19.0500 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 23.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1381 \cdot 23.8) / (\pi \cdot 1381^2)) } \\ &= 23.8120 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (23.8 - 19) / 23.8 \\ &= 0.2000 \end{aligned}$$

$$\begin{aligned} h/p &= 13.438601 ; \mu^* = 0.199983 ; E^* = 312334 \text{ kgf/cm}^2 \\ E^*/E &= 0.155329 ; \nu^* = 0.458596 \end{aligned}$$

Note: As h/p (13.439) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1400 + 88) 88) } \\ &= 0.0050 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\ &= 0.005 \cdot 2024848 \cdot 88^3 / (6(1 - 0.3^2)) \\ &= 1269582208.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned} &= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\ &= 6(1400 + 88) \cdot 1269582208 / 320^3 \cdot (1 + 320 \cdot 0.005 + (320 \cdot 0.005)^2 / 2) \\ &= 1348931.5 \text{ kgf/cm}^2 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned} &= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\ &= 1400^2 / (4 \cdot 2024848 \cdot 88) \cdot (1 - 1400 \cdot 0.3 / (2(1400 + 88))) \\ &= 0.23618E-02 \text{ mm/kgf/cm}^2 \end{aligned}$$

Parameter [w_s]:

$$\begin{aligned} &= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ &= 1.01 \cdot 1269582208 \cdot 0.005 \cdot 0.002361831 (1 + 320 \cdot 0.005) \\ &= 39815.9336 \text{ mm}^2 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned} &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1400 + 68) 68) } \\ &= 0.0058 \text{ 1/mm} \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\ &= 0.0058 \cdot 2041067 \cdot 68^3 / (6(1 - 0.3^2)) \end{aligned}$$



$$= 676284928.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned} &= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + \\ &\quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ &= 6(1400 + 68/\cos(0)) \cdot 676284928/320^3 \cdot (1 + \\ &\quad 320 \cdot 0.0058 \cdot (1 + 1)/2 + (320 \cdot 0.0058)^2/2 \cdot 1) \\ &= 824584.9 \text{ kgf/cm}^2 \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned} &= Dc^2/(4 \cdot Ec \cdot tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\ &= 1400^2/(4 \cdot 2041067 \cdot 68) \cdot (1 - 1400 \cdot 0.3/(2(1400 + 68))) \\ &= 0.30254E-02 \text{ mm/kgf/cm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot kc[\delta_c(CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \\ &\quad \zeta_c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)) \\ &= 1.01 \cdot 676284928 [0.003025411 (1 + 0.0058 \cdot 1 \cdot 320) - \\ &\quad 0.000 (1/320 + 1 \cdot 0.0058/2)) \\ &= 33903.8164 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 1576/1381 \\ &= 1.1411 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - v^*)/E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46)/312334 \cdot (1348931 + 824585 + \\ &\quad 2010787 \cdot \ln(1.14)) \\ &= 4.2277 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ &= 1381^2/16 \cdot [(1.01 - 1) \cdot (1.01^2 + 1) \cdot 157 - \\ &\quad (1.01 - 1) \cdot (1.01^2 + 1) \cdot 121] \\ &= 120217.9062 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\ &= 120218 + 33904 \cdot 121 - 39816 \cdot 157 \\ &= -2043076.6250 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\ &= ((-2043076) - 1381^2/32 \cdot 4.23 (157 - 121)) / (1 + 4.23) \\ &= -2142787.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2/64 \cdot (3 + v^*) \cdot (Ps - Pt) \\ &= -2142787 + 1381^2/64 \cdot (3 + 0.46) \cdot (157 - 121) \\ &= 1603516.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(| Mp |; | Mo |) \\ &= \max(| -2142787 |; | 1603516 |) \\ &= 2142787.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$



ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 2142787 / (0.2 (320 - 6)^2) \\ &= 652.0450 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1407 \\ &= 2814.26 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 2126194 / (0.2 (157 - 6)^2) \\ &= 2809.1819 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 157 + 0 + 0 \\ &= 156.6921 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(157 ; 55.7) \\ &= 156.6921 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$\begin{aligned} &= | P_s - P_t | \\ &= | 157 - 121 | \\ &= 36.344 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 320 / 1381 \cdot \min(0.8 \cdot 1407 ; 0.533 \cdot 2326) \\ &= 208.642 \text{ kgf/cm}^2 \end{aligned}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot | P_s - P_t | + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.2) \cdot (1381 / 1126) \cdot | 157 - 121 | + 0 \\ &= 55.7417 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 88.000mm
Min. Shell len. adjacent to the tubesheet is: 631.797 mm

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 1400^2 / (4 \cdot 88 (1400 + 88)) \cdot 157 \\ &= 589.3330 \text{ kgf/cm}^2 \end{aligned}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P's + 6(1 - \nu) / E \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 27989480 / 88^2 \cdot [0.005 \cdot 0.37 + 6(1 - 0.46) / 312334 \cdot \\ &\quad 1381 / 320^3 \cdot (1 + 320 \cdot 0.005 / 2) \cdot (-2142787 + 1381^2 / 32 \cdot \\ &\quad (157 - 121))] \\ &= 1856.399 \text{ kgf/cm}^2 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned} &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq 1.5 \cdot S_s \text{)} \\ &= 2445.732 \text{ kgf/cm}^2 \text{ (Should be } \leq 2446.49 \text{)} \end{aligned}$$

Note: For a given Channel thickness of: 68.000 mm



Min. Channel len. adjacent to the tubesheet is: 555.381mm

Axial Channel Membrane Stress [σ_c, m]:
 $= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
 $= 1400^2 / (4 \cdot 68 (1400 + 68) \cdot \cos(0)) \cdot 121$
 $= 594.657 \text{ kgf/cm}^2$

Axial Channel Bending Stress [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu) / E) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
 $= 6 \cdot 14909521 / 68^2 \cdot [(0.003025411 \cdot 1 \cdot 0.0058 - 0.000 \cdot 1) \cdot 121 - 6((1 - 0.46) / 312334) \cdot 1381 / 320^3 \cdot (1 + 0.0058 \cdot 320 / 2 \cdot 1) \cdot (-2142787 + 1381^2 / 32 \cdot (157 - 121))]$
 $= 1833.084 \text{ kgf/cm}^2$

Channel Membrane + Bending Stress [σ_c]:
 $= | \sigma_{c,m} | + | \sigma_{c,b} |$ (Should be $\leq 1.5 \cdot S_c$)
 $= 2427.741 \text{ kgf/cm}^2$ (Should be ≤ 2507.16)

Tube Metal Area [A_t]:
 $= \pi (d_o - t) t$
 $= \pi (19 - 3.05) 3.05$
 $= 1.532 \text{ cm}^2$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 3.05 (19 - 3.05) 942$
 $= 1443.667 \text{ kgf}$

Groove Weld Strength [F_g]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not $> F_t$)
 $= 0.85 \cdot \pi \cdot 3.05 (19 + 0.67 \cdot 3.05) 942$
 $= 1443.6666 \text{ kgf}$

Weld Strength Factor [f_w]:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 942 / 942$
 $= 1.000$

Design Strength Ratio [f_d]:
 $= 1$ for full strength welds

Fillet Weld Strength [F_f]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not $> F_t$)
 $= 0.55 \cdot \pi \cdot 3.05 (19 + 0.67 \cdot 3.05) 942$
 $= 1046.5896 \text{ kgf}$

Minimum Length of Weld [a_r]:
 $= 2 \cdot \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o}$
 $= 2 \cdot \sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1) - 0.75 \cdot 19}$
 $= 3.445 \text{ mm}$

Minimum Required Fillet Weld Leg [a_{fr}]: 1.8288 mm
Minimum Required Groove Weld Leg [a_{gr}]: 1.8288 mm

Tube-Tubesheet Jt allowable, 1444 is $\geq$ tube strength 1444 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	652.0 <=	2814.3 kgf/cm ²	Ok
Tubesheet shear stress	196.1 <=	1125.7 kgf/cm ²	Ok
Stress in Shell at Tubesheet	2445.7 <=	2446.5 kgf/cm ²	Ok



Stress in Channel at Tubesheet 2427.7 <= 2507.2 kgf/cm² Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	156.692	320.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	3.048	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	3.048	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	272.049	...	1988	2814	654 1126	Ps+Pt D1 Ok
D2uc	310.541	...	2636	2814	850 1126	Ps+Pt D2 Ok
D3uc	156.692	...	652	2814	196 1126	Ps+Pt D3 Ok
D1c	273.877	...	2003	2814	667 1126	Ps+Pt-c D1 Ok
D2c	312.115	...	2656	2814	867 1126	Ps+Pt-c D2 Ok
D3c	159.554	...	661	2814	200 1126	Ps+Pt-c D3 Ok
Max:	312.1145	... mm	0.944	0.770	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Shell	Reqd. Thk. Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	523	2446	2942	5015	Ok
D2uc	...	...	2969	4901	514	2507	Ok
D3uc	...	...	2446	2446	2428	2507	Ok
D1c	...	...	604	2446	3154	5015	Ok
D2c	...	...	3170	4901	601	2507	Ok
D3c	...	...	2566	4901	2553	5015	Ok
Max	...	...	1.000	0.968			

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	312.115	320.000	Ok
Tube thickness :	1.751	3.048	Ok
Tube-tubesheet fillet weld leg :	1.829	3.048	Ok
Tube-tubesheet groove weld leg :	1.829	3.048	Ok



Given Shell Thickness: 88.0000 mm
 Given Channel Thickness: 68.0000 mm

Min Shell length of thk, (88.000) adj. to tubesheet: 631.797 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 555.381 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	169.647	1.000	165.970	1.000
Tubesheet Shear Stress	204.566	1.000	204.566	1.000
Shell Stress (Axial, Junction)	204.566	0.417	208.641	1.257
Tubesheet-Channel Junction	169.647	1.000	208.641	1.257
Tube Pressure Stress	363.358	1.000	334.554	1.000
Minimum MAWP	169.647		165.970	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	171.122	1.000	167.607	1.000
Tubesheet Shear Stress	208.641	1.000	208.641	1.000
Shell Stress (Axial, Junction)	208.641	1.219	208.641	1.245
Tubesheet-Channel Junction	208.641	1.219	208.641	1.245
Tube Pressure Stress	363.358	1.000	-0.001	1.000
Minimum MAPnc	171.122		-0.001	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(320/4, \min(320, 88))$$

$$= 88.000 \text{ mm}$$

Note: This Element/Detail was specified as being Impact Tested.

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(320/4, \min(320, 68))$$

$$= 80.000 \text{ mm}$$

MDMT reduction ratio:

$$= \min(\text{pt/Tubeside MAPnc}, 1)$$

$$= \min(121/171, 1)$$

$$= 0.708$$

Note: This Element/Detail was specified as being Impact Tested.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: